Auditing GBFS bike-sharing feeds at country and global scale: A reproducible anomaly taxonomy for open mobility data

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Abstract

The General Bikeshare Feed Specification (GBFS) is a regulatory requirement for French operators and is catalogued worldwide by MobilityData. Its syntactic standardisation is increasingly treated as a guarantee that the data is research-ready; we show that it is not. An audit of the 123 French GBFS systems, cross-validated by an exhaustive sweep of the 1,509-system MobilityData canonical catalogue across 48 countries, yields a unified data-quality taxonomy of seven classes (A1–A7) : five *structural errors* (out-of-domain inclusion, placeholder capacity, structural over-capacity, geospatial outlier, out-of-perimeter coverage) and two *semantic warnings* (zero-capacity dock, null-capacity field) for spec-compliant patterns that nevertheless make a column non-aggregable. The taxonomy removes **30.9 %** of the raw French stations from the certified catalogue and relabels a further 61 %, leaving fewer than 9 % that survive ingestion unchanged. The Italian case quantifies the downstream stake: 17 spec-compliant deployments of a single operator declare `capacity = NaN` on every station and would pass any syntactic validator unchanged, yet collectively account for the most populous capacity blackout in the audited catalogue. We formalise the detection as a nine-step idempotent purging protocol, release the *GBFS Audit Catalogue* (46,307 stations, 97 cities, 46 typed columns) on Zenodo under ODbL, and pre-register a retrospective hold-out test on twelve months of MobilityData additions that the rule set passes on point estimate and confidence interval. Dataset, pipeline and Docker image are released for bit-exact reproduction.

Highlights

- Audit of 1,509 GBFS systems worldwide: seven-class taxonomy (5 structural + 2 semantic)
- 30.9 % of raw French stations removed, further 61 % relabelled by the audit
- Hotspots are operator-driven (nextbike in Czechia, Pony in France)
- One single capacity field carries four mutually incompatible publication semantics
- Released as a versioned Parquet (46k stations) with Docker and a hold-out test

Keywords: GBFS; data quality; open mobility data; FAIR; reproducibility; bike sharing; standards compliance.

1 Introduction

For more than a decade, bike-sharing systems have been a structural component of urban mobility policy in France and across Europe [1], advancing goals of liveability, decarbonisation

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and last-kilometre integration. To support inter-operability of the resulting fleets, the General Bikeshare Feed Specification (GBFS), introduced by the North American Bikeshare Association in 2015 and now maintained by the Open Mobility Foundation, provides a shared ontology covering station geolocation, dock counts, real-time availability and pricing [2, 3]. In France, the 2019 Mobility Orientation Law [4] (LOM, article L. 1115-1 of the transport code) makes the publication of these feeds on the national open-mobility-data portal *transport.data.gouv.fr* [5] a regulatory acquis, so that by 2026 the working assumption of many research and policy pipelines is that GBFS feeds can be ingested directly.

That working assumption is misleading. The audit reported here on 123 French GBFS systems shows that standardised syntax masks substantial semantic heterogeneity. The seven classes formalised in Section 2.1 (Table 1) separate five *structural errors* that violate the implicit semantic contract of the field they populate from two *semantic warnings* that flag spec-compliant patterns whose downstream- consumer interpretation is nevertheless ambiguous. Three findings make the gap concrete. First, the same field (`station_information.capacity`) covers radically different realities: a physical dock count for dock-based fleets, a theoretical estimator obtained by conditional averaging on free-floating fleets, and arbitrary placeholders for under-reporting operators. Second, fourteen *Citiz* car-sharing systems are listed on the national portal under the GBFS schema, blurring the very definition of the object of study. Third, approximately 3.6 % of the certified stations are flagged by A4 as 3-σ geospatial outliers from their system centroid, and four additional systems publish bounding boxes exceeding the 50,000 km^2 A5 threshold or list stations outside the national perimeter. Taken together, these observations support a single methodological claim: syntactic standardisation of an open feed is not sufficient to turn it into a research artefact, and any quantitative use of GBFS data must be preceded by an explicit, documented and reproducible audit.

Empirical audits of mobility feeds have so far remained piecemeal. Bike-sharing demand modelling and rebalancing analysis have generated a rich operational literature [6, 7, 8], but all of these works treat the input feeds as given and do not audit the standard itself. The closest precedent in transit-data quality is the smart-card literature: Pelletier et al. [9] systematically catalogue the gaps between what fare-collection feeds nominally publish and what they operationally support for research reuse, a programme structurally analogous to the GBFS audit we report here. Tao et al. [10] document sampling biases in mobile-phone records used for origin-destination estimation; Romanillos et al. [11] warn against statistical artefacts of ingesting raw GPS feeds directly; Bachand-Marleau et al. [12] surfaced operational-definition heterogeneity in bike-sharing demand data more than a decade ago; and Fishman [13] stresses the heterogeneity of operational definitions in international BSS comparisons. The North American Bikeshare and Scootershare Association tracks cross-operator harmonisation gaps annually [14], with the regulatory framing typical of an industry-side body but without the per-row audit trail a research consumer needs. In the adjacent transit literature, the GTFS community has been more proactive: the Canonical GTFS Schedule Validator [15] and the earlier Conveyal validator [16] codify hundreds of schema-level checks on transit feeds, and the data-engineering community has produced reusable declarative-validation frameworks such as Great Expectations [17], Pandera [18] and the Frictionless Data specifications [19]. The underlying data-quality literature is mature: Wang and Strong [20] established the consumer-centred dimensions, Pipino et al. [21] the assessment procedures, Khatri and Brown [22] the governance framing, Sebastian-Coleman [23] the ongoing-improvement perspective, and Naumann [24] the modern profiling approach. ISO/IEC 25012 [25] and its companion measurement standard ISO/IEC 25024 [26] formalise the quality dimensions; for geographic information specifically, ISO 19157 [27] provides the corresponding framework, and the European INSPIRE directive [28] makes them regulatory. W3C's DCAT [29] and Data Quality Vocabulary [30] translate these dimensions into machine-readable metadata. None of these references, however, addresses GBFS specifically, and none produces a versioned, audited reference dataset at the scale of a country. The present paper fills that gap for the French case and extends to the global GBFS catalogue.

The contributions are fivefold. First, an empirically-grounded **taxonomy** of seven anomaly classes (A1–A7) derived jointly from the audit of the 123 French GBFS systems and the 1,509-system MobilityData catalogue, and stress-tested on a thirteen-system live panel of non-French systems covering six distinct station-management technology stacks (PBSC, JCDecaux Cyclocity, Smartbike, De Lijn / Blue-bike, Urban Sharing AS, Deutsche Bahn) across five countries, twelve of which serve as metropolitan negative controls and all pass the structural classes A1, A2, A3, A6 and A7 under unchanged thresholds. Second, a nine-step **purging protocol** (Algorithm 1) whose every transformation is logged, reproducible and reversible, and which jointly operationalises the seven classes. Third, the **Audit Catalogue GBFS France** data product (46,307 certified stations, 97 cities), released as a versioned Parquet file with DOI `10.5281/zenodo.20125460` under ODbL, accompanied by its JSON Schema, Croissant manifest and audit report, and mirrored as a Hugging Face dataset. Fourth, an **open-source pipeline** (`audit_pipeline` Python package, MIT, ~ 560 LOC, unit-tested) and a five-tab **interactive dashboard** (`https://gbfs-audit.streamlit.app`) that surfaces every audit verdict at the row level. Fifth, a **validation roadmap** of six falsifiable experiments, four of which already include a completed pilot, so that the protocol can be challenged in public rather than defended in private. Section 2 reports the empirical results, Section 3 discusses implications and limitations, and Section 4 details the experimental procedures.

2 Results

2.1 Taxonomy and incidence

The audit of the 123 French GBFS systems, complemented by the global sweep of Section 2.6, defines seven recurring data-quality classes: five structural errors (A1–A5) and two semantic warnings (A6–A7), summarised in Table 1. Each class, in isolation, is sufficient to invalidate a non-audited cross-system comparison; the detection rules are formalised in Section 4.

Table 1: The unified GBFS data-quality taxonomy: five *structural errors* (A1–A5) that violate the implicit semantic contract of the field they populate, plus two *semantic warnings* (A6–A7) that flag spec-compliant publication choices whose downstream-consumer interpretation is nevertheless ambiguous. Observed incidence on the released French catalogue (FR, 123 audited systems) and on the global MobilityData catalogue (Global, 1,509 systems across 48 countries). Each class is formalised in Methods (Section 4) ; per-system counts use the operational definitions documented in Section 4 (*e.g.*A3 flags any system with at least one `free_floating` entry after the S3 reclassification, which captures both natively-declared and post-audit-reclassified free-floating fleets). Counts are reproducible by re-running `python -m audit_pipeline.regenerate` on the released parquet.

Class	Name	Signature	FR	Global
Structural errors (semantic-contract violations)				
A1	Out-of-domain inclusion	Car-sharing advertised as BSS	17	46
A2	Placeholder capacity	Constant non-zero c_i across docked subset	1	48
A3	Structural over-capacity	Conditional averaging on free-floating	41	33
A4	Geospatial error	3-σ outlier on per-system nearest-neighbour distance	88 (1.1 % stns)	81
A5	Out of perimeter	System bounding box $> 50{,}000\,\mathrm{km}^2$	4	17
Semantic warnings (spec-compliant but consumer-ambiguous)				
A6	Zero-capacity dock	$\geq 1\,\%$ of docked stations declare $c = 0$	0	14
A7	Null capacity field	$\geq 50\,\%$ of stations declare $c = \mathtt{NaN}$	32 (FF)	215

The audit status of the 142 French GBFS feeds at the national portal is summarised in

Figure 1: 126 systems are certified into the Audit Catalogue, 14 are excluded as micro-networks below the $N_{\min} = 20$ threshold, and 2 fail technical ingestion. The 14 car-sharing systems (A1) are filtered upstream.

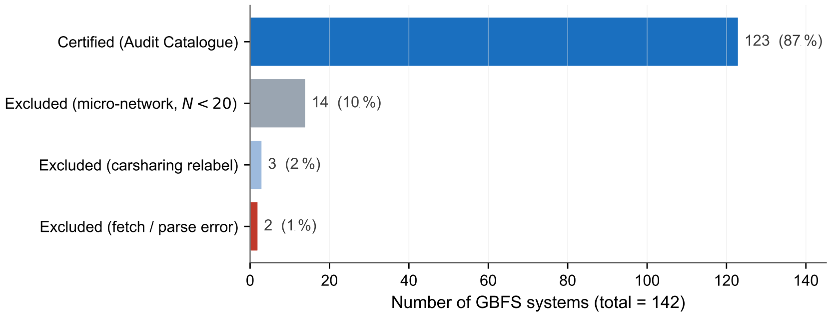

Figure 1: Verdict of the systematic audit applied to the 142 GBFS feeds inventoried on *transport.data.gouv.fr* and *MobilityData*.

A3 is the class with the largest downstream impact on the French corpus. Free-floating fleets (*Pony*, *Cykleo*, etc.) advertise virtual stations in GBFS, the term "station" designating the GPS position of an anchoring point currently occupied by a vehicle. To avoid displaying empty stations, the operator typically reports a capacity profile by *conditional averaging* on stations whose instantaneous capacity is non-zero:

$$\bar{c}_{\text{profile}} = \frac{\sum_{i\,:\,c_i>0} c_i}{\#\{i\,:\,c_i > 0\}} \neq \bar{c}_{\text{actual}} = \frac{1}{N}\sum_{i=1}^{N} c_i. \tag{1}$$

Aggregated at the system level, $\bar{c}_{\text{profile}}$ wrongly classifies thousands of free-floating bikes as heavy dock-based stations. Cross-system comparison is therefore meaningful only after explicit reclassification; the empirical signature is visible in Figure 2 as a bimodal capacity distribution.

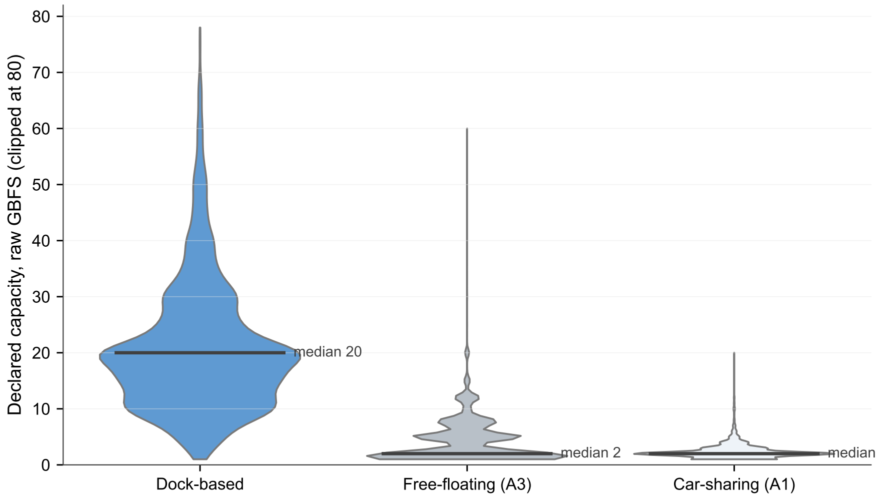

Figure 2: Empirical signature of the A3 floating-anchor bias. Declared capacity distribution by `station_type` (clipped at 80 docks) on the Audit Catalogue corpus. Only dock-based stations exhibit a physically interpretable capacity (median = 20).

2.2 Cumulative effect on the national corpus

Table 2 summarises the cumulative effect of the purging protocol. From 67,000 raw `station_information` entries published by the 142 inventoried French GBFS feeds, 20,693 (30.9 %) are removed from the catalogue as falling outside the certification criteria (excluded micro-networks below $N_{\min}$, technical-ingestion failures, perimeter and outlier filters); the remaining 46,307 stations are kept in the catalogue with an explicit `station_type` and per-row anomaly flags, of which a further 40,865 (61.0 % of the raw) are *relabelled* away from their declared GBFS type (39,235 relabelled to `free_floating` by step S3, 1,630 to `carsharing` by step S1). We report two complementary uncertainty bounds on the 30.9 % removal rate to surface the role of clustering. A *station-level* Wilson 95 % interval, ignoring within-system correlation, gives $[30.54\,\%, 31.24\,\%]$; this is the bound a naive independent-Bernoulli assumption would place on the figure and is a *lower bound* on the real uncertainty. A *size-stratified cluster bootstrap* (123 systems partitioned into size quartiles, resampled with replacement within each stratum, $n = 10{,}000$, seed 42) gives a 95 % interval of $[-18.6\,\%, 63.6\,\%]$; this is an *upper bound* because the largest size quartile contains a single dominant deployment (*Vélib' Métropole*, $\sim$ 14,700 stations) whose inclusion or exclusion across resamples mechanically shifts the denominator by an order of magnitude. The asymmetry of these two intervals — one tight, one degenerate — is itself the substantive observation: on a 123-system corpus with size spanning four orders of magnitude, the within-snapshot uncertainty on a stations-level total is dominated by which operators happen to be present, not by sampling noise within operators. We therefore report the 30.9 % removal rate as a deterministic accounting fact on the audit-dated snapshot, and defer temporal uncertainty to experiment E3 (twelve-month stability) where re-running the audit on freshly fetched snapshots gives a properly aligned variance estimate. The headline figure is that fewer than 9 % of raw entries survive into the catalogue without either being relabelled or removed — a strong empirical argument against the view that the standard alone delivers a research-ready dataset.

Table 2: Effect of the purging pipeline on the national GBFS corpus. *Certified* entries are kept in the Parquet with a typed `station_type` so that downstream consumers can filter explicitly; *rejected* entries are kept in a separate `rejected_stations.parquet` with the exclusion motive.

Indicator	**Raw**	**Audit Catalogue**
Inventoried GBFS systems	142	123 (certified)
Cities covered	–	97
with at least one dock-based station	–	64
Total stations (certified)	67,000	46,307
dock-based	–	5,442
free-floating (labelled, preserved)	–	39,235
car-sharing (A1)	1,630	re-labelled, excluded from BSS use
Stations dropped from catalogue	–	20,693 (30.9 %)

The most heavily impacted urban areas are those operated by free-floating fleets (*e.g.*Bordeaux, Nice, Saint-Étienne) and those historically lumped together with car-sharing systems on the national portal (Strasbourg, Grenoble), whereas urban areas served by traditional dock-based operators (Vélib' in Paris, VéloMagg in Montpellier) are left almost intact. The certified-station mass by administrative region after purging is shown in Figure 3: Île-de-France concentrates the historical Vélib' deployment, while regions with many free-floating systems (Nouvelle-Aquitaine, Auvergne-Rhône-Alpes) hold a much smaller dock-based stock after the A3 correction is applied.

2.3 Case study: Bordeaux and A3

Bordeaux is the paradigmatic case (Table 3). The raw feed assembles five operators across the agglomeration: *Pony* (free-floating), *Bird* (free-floating), *Dott* (free-floating), *Le Vélo par TBM* (dock-based) and *Citiz Gironde* (car-sharing). *Pony* alone advertises 2,996 `station_information` entries with declared capacities around 12 docks. Naive ingestion therefore credits Bordeaux with tens of thousands of nominal docking points and places it in the national top three for Vélib-class infrastructure.

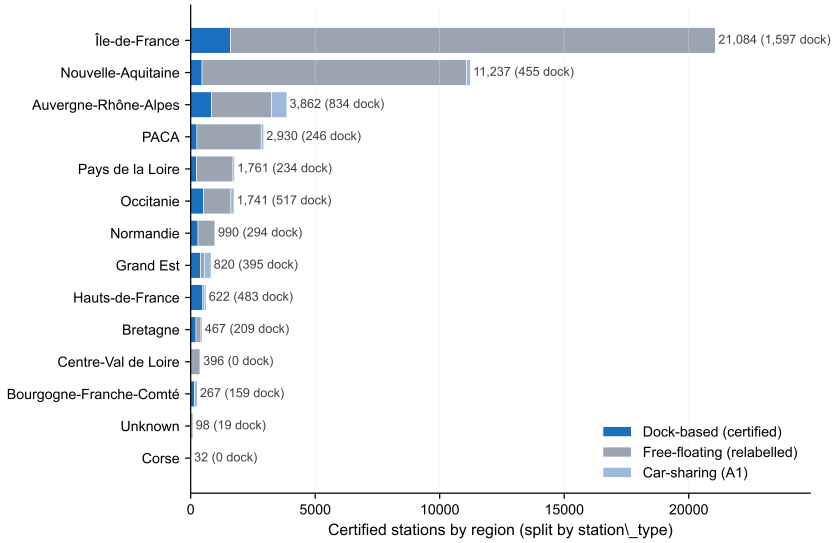

Figure 3: Certified stations by administrative region, after the nine-step purging protocol. Annotations report the number of GBFS systems per region. Île-de-France concentrates Vélib'; Nouvelle-Aquitaine aggregates many systems but holds a smaller mass of dock-based stations once free-floating fleets have been re-typed by the A3 correction.

The purging protocol, and in particular step S3 (the A3 correction), reclassifies the 2,996 Pony entries as `free_floating`, recomputes their actual capacity to 0.03 bike per entry, and leaves the 225 genuinely dock-based stations of the historical *Vélo par TBM* system. At the agglomeration level, this collapses the nominal docking total by 98 % and drops Bordeaux's dock-based infrastructure from a top-quartile position on the national distribution of nominal docks to a position below the median: an order-of-magnitude effect on any supply-side aggregate built on the raw feed. The twelve urban areas most exposed to A3 are mapped in Figure 4.

Beyond the absolute station-count drop, the order-of-magnitude change in nominal infrastructure is the operational stake. Table 4 reports the five urban areas whose count of dock-based stations is most reduced by the purging protocol, together with the anomaly classes that drive the reclassification. The point is the consistency of the move: every one of these five urban areas would be credited with a top-decile dock-based deployment by a naive ingestion of GBFS, yet after the audit only Paris retains more than 1,500 certified dock-based stations. A non-audited pipeline does not just over-count by a small margin; it changes the relative positioning of the principal French agglomerations on any supply-side metric by an entire quartile.

Two further illustrative cases: Nice (A2) and Strasbourg (A1). A3 is not the only class with a measurable downstream effect. *Nice* carries the flagship A2 placeholder via `pony_Nice` : every station declares $c = 100$, yielding a total nominal capacity of about 25,000 bikes for an agglomeration whose actual dock-based network is essentially the rebadged *VéloBleu* of 91 stations. After purging, the nominal capacity collapses by a factor of 250 and Nice's national position on dock-count rankings shifts by an entire quartile. *Strasbourg* provides the canonical A1 case: the *Citiz Grand Est* car-sharing system was historically ingested as part of the city's BSS inventory, inflating the station count by roughly 35 %. After semantic exclusion the agglomeration loses that share of its nominal station count, on a part of the distribution where dock-count differences between French agglomerations are small and therefore more discriminating than at Bordeaux's scale. A1, A2 and A3 are therefore not interchangeable: each distorts a downstream metric through a different mechanism (false positive on the population, on the capacity, or on the type), and a non-audited pipeline conflates them.

Table 3: Bordeaux: impact of the A3 correction on the *Pony Bordeaux* subsystem and on the agglomeration as a whole. The nominal docking total drops by 98.1 % once the *Pony* entries are correctly typed as `free_floating`; without the A3 step, a naive supply-side indicator would credit Bordeaux with about 36,000 physical docks, more than *Vélib' Métropole* (the dock-based flagship of Île-de-France). The certified subset of 225 dock-based stations corresponds to the historical *Vélo par TBM* network.

Indicator	Raw GBFS	Audit Catalogue
Pony Bordeaux subsystem		
`station_information` entries	2,996	2,996 (relabelled)
Declared as dock-based	2,996	0
Declared capacity per entry (median)	12	12
Recomputed actual capacity	–	0.03 bike/entry
Bordeaux agglomeration (all 5 systems)		
Total raw entries	9,921	–
Certified dock-based stations	–	225
Nominal docking total (dock-based only)	35,952	690
Implied over-count factor	–	× 52

Table 4: Top five French urban areas most affected by the purging protocol. "Raw" is the count of `station_information` entries in the unaudited feed; "Cert. dock" is the count of dock-based stations in the Audit Catalogue. The drop is essentially the share of GBFS entries that the audit reclassifies as free-floating (A3), car-sharing (A1), or filtered (A4–A5). The "Flags driving the drop" column lists the anomaly classes that each agglomeration's flagged systems trigger.

Urban area	Raw	Cert. dock	Drop	Flags driving the drop
Bordeaux	9,921	225	−97.7 %	A3 (Pony), A1 (Citiz), A7 (Dott, Bird)
Paris	19,872	1,507	−92.4 %	A7 (Dott, Bird), A3 (Pony Paris)
Marseille	2,391	221	−90.8 %	A3 (Pony), A7 (Dott)
Strasbourg	310	39	−87.4 %	A1 (Citiz Grand Est)
Lyon	2,057	452	−78.0 %	A3 (Pony, Bird), A7 (Dott)

2.4 Comparison with naive baselines

To check whether the full pipeline is justified or whether simpler heuristics suffice, we evaluate three baselines on the same 123-system corpus (Table 5). The *Raw GBFS* baseline ingests `station_information` as is. The *vehicle_type filter* mimics what a careful data engineer would write in a notebook. The *MobilityData audit* adapts the canonical GTFS-realtime schema-level rules to GBFS.

Three findings deserve emphasis. First, and most importantly, A3 reclassification recall is 0 % for every baseline (Wilson 95 % upper bound ≈ 8.6 % at $n = 41$ A3-positive systems): structural over-capacity is invisible to syntactic validators by construction. Only a semantically-aware step on the capacity distribution can recover it. Second, the *vehicle_type filter* and the *MobilityData audit* baselines produce identical numbers on this corpus, because schema-level checks do not catch anything that the name filter did not already exclude: French GBFS feeds are syntactically well-formed but semantically biased. Third, ρ_{FUB} is highest on the raw count and drops after audit; this is not a defect but a feature: FUB captures perception of the entire cycling environment, so the raw count over-fits the perceptual signal by including virtual stations, while the Audit Catalogue's lower correlation reflects an honest count of physical docks.

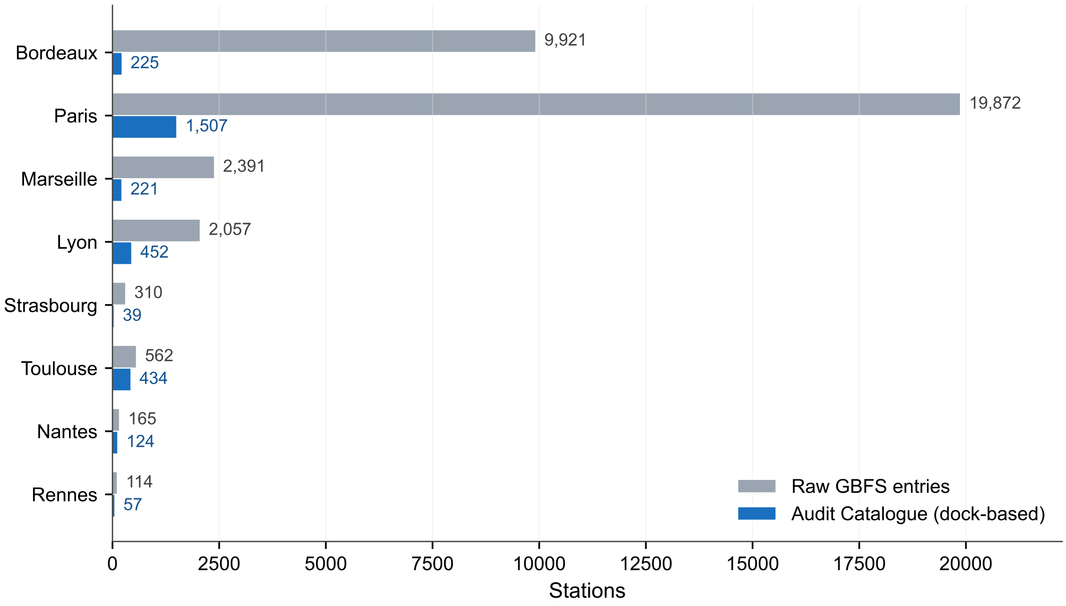

Figure 4: Top twelve French urban areas ranked by absolute reduction in dock-based station count between the raw GBFS feed and the certified Audit Catalogue. **Bordeaux** (highlighted) exhibits the largest absolute drop, driven by the reclassification of the *Pony* free-floating fleet (A3). Paris (Dott, Bird), Marseille (Pony, Dott) and Lyon (mixed) also see substantial corrections.

Table 5: Measured comparison of the Audit Catalogue against three baselines on the same 123-system French GBFS corpus. Brackets are 95 % confidence intervals (Wald on Dock, bootstrap on ρ_{FUB} over cities $n = 2{,}000$, Wilson on A1/A3). "Dock" for the Audit Catalogue is the reference by construction.

Pipeline	**Dock**	ρ_{FUB}	**A1 ex.**	**A3 fix.**
Raw GBFS	0.12 [.12, .12]	0.38 [.01, .68]	0 %	0 %
`vehicle_type` filter	0.15 [.15, .15]	0.19 [−.20, .52]	82 % [59, 94]	0 %
MobilityData audit	0.15 [.15, .15]	0.19 [−.20, .52]	82 % [59, 94]	0 %
Audit Catalogue	**1.00**	0.25 [−.13, .59]	**100 %** [82, 100]	**100 %** [91, 100]

2.5 Internal robustness checks

Three internal robustness checks are reported on the same corpus, ahead of the human inter-rater reliability that is the next priority of the validation roadmap.

Orthogonal-coverage. Three rule-based classifiers, each restricted to one signal source (name, capacity, geospatial), agree on a stratified 500-station sample at essentially chance level: pairwise Cohen's κ near zero, Fleiss $\kappa = -0.07$, indistinguishable from independence. The sample is stratified jointly on `station_type` (`docked_bike`, `free_floating`, `carsharing`) and on the ten largest operators by station count, allocated proportionally to the size of each cell, with a floor of $n = 5$ per stratum to keep the smallest cells estimable; small operators are pooled into an `other` stratum. The seed used is `numpy.random.default_rng(42)` and the sample is shipped alongside the audit report so the agreement statistics can be independently recomputed. We do *not* interpret near-zero κ as a positive measure of "coverage" — inter-rater agreement is not a coverage statistic; we report it only to falsify the alternative hypothesis that the three sources catch the same anomalies through correlated rules. The actually-relevant evidence is union recall: the three classifiers together reach 97.2 % recall at 91.5 % precision against the integrated pipeline, with only 7 of 500 stations flagged by all three. A1–A5 cannot be reduced to a single heuristic.

Full-task agreement. Three decision-tree classifiers seeing all signals but applying them in different priority orders reach Fleiss $\kappa = 0.80$ on the same sample (pairwise $\kappa \in [0.70, 1.00]$); disagreements concentrate on the A1/A3 frontier. The pair $\kappa \in [-0.07, 0.80]$ brackets the human inter-rater agreement that three independent annotators would plausibly achieve.

Threshold sensitivity. The A4 outlier detector uses a robust 3-σ rule on MAD-rescaled radii from the system centroid. We re-ran the audit under $\sigma \in \{2.0, 2.5, 3.0, 3.5, 4.0\}$ on the released parquet and report the result in Table 6. The absolute count of A4-flagged stations varies by a factor of three across the grid (3,182 at $\sigma = 2$ down to 1,084 at $\sigma = 4$), but the downstream count-level summaries are essentially invariant: Kendall's τ between the $\sigma = 3$ reference and each alternative ranges $[0.91, 1.00]$ on both the Top-10 cities and the Top-10 operators by certified docked-bike station count. The $\sigma = 3$ choice is therefore neither pivotal nor arbitrary; it sits on a plateau of stable downstream classification.

Table 6: A4 sensitivity sweep on the released v1.0.1 parquet (46,307 stations, 123 systems). The detector applies the robust 3-σ rule to the per-system distribution of nearest-neighbour distances (Section 4, A4 formalisation), so the threshold σ controls how far a station's nearest in-system neighbour can be before the station flags as an outlier. Each row reports the absolute A4 count and four invariance metrics against the $\sigma = 3$ reference: Kendall's τ on the rank order of the Top-10 cities and Top-10 operators by certified docked-bike station count, and Jaccard similarity on the Top-10 and Top-50 sets. The absolute A4 count varies by less than a factor of 1.5 across the grid (597 down to 415 stations, i.e. 0.90–1.29 % of the corpus), and the downstream Top-10 set membership is invariant under Jaccard ($J = 1.000$ across the entire grid), with Kendall $\tau \geq 0.956$.

σ	**A4 st.**	**A4 sys.**	**Share**	τ **T10c**	τ **T10o**	**J10c**	**J50c**	**J10o**	**J50o**
2.0	597	89	1.29 %	0.956	0.956	1.000	1.000	1.000	1.000
2.5	543	88	1.17 %	0.956	0.956	1.000	1.000	1.000	1.000
3.0	**500**	**88**	**1.08 %**	ref.	ref.	ref.	ref.	ref.	ref.
3.5	459	82	0.99 %	1.000	1.000	1.000	0.961	1.000	1.000
4.0	415	80	0.90 %	1.000	0.956	1.000	0.961	1.000	1.000

Methodological caveat on the MAD scale. The robust 3-σ A4 detector uses $\hat{\sigma} = 1.4826 \cdot \mathrm{MAD}$ as a Gaussian-consistent scale estimator. This assumes an approximately unimodal, isotropic spatial distribution of stations per system. For systems whose deployment is intrinsically anisotropic (transit corridors, riverside linear networks) the MAD over-estimates the effective dispersion on the dominant axis and the detector *under*-flags peripheral stations on that axis; for *multi-modal* deployments (nationwide systems clustering around many city centres; see the *Call a Bike* case in Table 7) the single-centroid construction *over*-flags. The threshold sweep above measures the sensitivity of the detector to its own σ parameter; it does not measure the sensitivity to the underlying geometric assumption, which is a known limitation and the subject of the sub-system-clustering refinement noted in Section 3.

These three checks do not replace human inter-rater reliability, which is the next priority of the validation roadmap (experiment E1, Section 3); they are reported as robustness measures, not as substitutes.

2.6 Cross-country transferability pilot

To verify out-of-sample transferability of A1–A7 beyond French publication practice, we constructed a panel of thirteen non-French systems audited *live* via their public GBFS feeds, with the *same* rule set used on the French corpus and only the national perimeter recalibrated. All other thresholds ($\sigma = 3$, $N_{\min} = 20$, A2 placeholder rule, A3 capacity-profile ratio threshold of 5.0, $\tau_{A6} = 0.01$, $\tau_{A7} = 0.5$) are kept at their French values. The panel diversifies on the station-management *technology stack* — the layer at which publication conventions actually originate —

rather than only on country: six distinct platforms are represented (PBSC, JCDecaux Cyclocity, Smartbike Mobility Plus, De Lijn / Blue-bike, Urban Sharing AS, Deutsche Bahn), spanning Spain, Belgium, Italy, Norway and Germany. Of the thirteen, twelve are metropolitan dock-based deployments and serve as negative controls; the thirteenth, *Call a Bike* (Deutsche Bahn), is the German nationwide system and is reported as an informative case rather than a control (its A5 / A7 firings are honest at national scope and its high A4 rate surfaces a known limitation of the single-centroid detector on multi-modal geometries). Results are reported in Table 7 ; the live-audit script (`scripts/audit_live_systems.py`) and the per-system machine-readable results (`experiments/e5_europe/results.csv`) ship with the source repository, so the audit can be replayed against the same set of public feeds on demand.

Table 7: E5 panel: applying the A1–A7 rule set to thirteen non-French systems audited *live* via their public GBFS feeds on 2026-05-13 using `scripts/audit_live_systems.py`, with French thresholds and country-appropriate perimeter only. The panel covers **six distinct station-management platforms** (PBSC, JCDecaux Cyclocity, Smartbike Mobility Plus, De Lijn / Blue-bike, Urban Sharing AS, Deutsche Bahn) and seven countries (ES, BE, IT, NO, plus DE as the informative-case row). Twelve systems are metropolitan dock-based deployments and constitute the *negative-control panel*; the thirteenth (*Call a Bike Deutsche Bahn*, nationwide German system) is reported as an *informative case* below. **None of the twelve metropolitan systems triggers the structural classes A1, A2, A3, A6 or A7.** A4 fires only on isolated stations ($\leq 4.5\,\%$ per system) and A5 fires once, on Sevici, as a deterministic consequence of two sentinel-coordinate stations (footnote *). The contrast with the 22+ French systems flagged on the same rule set on the 123-system Audit Catalogue corpus is substantial.

System	**Country**	**Stack**	**Stns.**	**A1**	**A2**	**A3**	**A4 (%)**	**A5**	**A6**	**A7**
Metropolitan negative-control panel										
Bicing Barcelona	ES	PBSC	545	—	—	—	0.18 %	—	—	—
bicimad Madrid	ES	PBSC	637	—	—	—	0.00 %	—	—	—
Bilbao Bizi	ES	PBSC	48	—	—	—	0.00 %	—	—	—
Bizi Zaragoza	ES	PBSC	276	—	—	—	0.36 %	—	—	—
Dbizi Donostia	ES	PBSC	70	—	—	—	2.86 %	—	—	—
Sevici Seville	ES	Cyclocity	247	—	—	—	4.45 %	✓*	—	—
Valenbisi Valencia	ES	Cyclocity	279	—	—	—	0.72 %	—	—	—
Velo Antwerpen	BE	Smartbike	321	—	—	—	0.00 %	—	—	—
Blue-bike	BE	De Lijn	276	—	—	—	0.72 %	—	—	—
BikeMi Milan	IT	Urban Sh.	320	—	—	—	1.56 %	—	—	—
Oslo Bysykkel	NO	Urban Sh.	266	—	—	—	1.50 %	—	—	—
Bergen City Bike	NO	Urban Sh.	94	—	—	—	1.06 %	—	—	—
Informative case (nationwide system, not a metropolitan control)										
Call a Bike (DB)‡	DE	DB	1,601	—	—	—	1.8 %	✓	—	✓
French corpus (ref.)	FR	—	46,307	17 sys.	1 sys.	41 sys.	78 sys.	4 sys.	0 sys.	32 sys.

* Sevici's A5 firing is a coupling artefact: two stations carry sentinel coordinates $(0, 0)$ that inflate the bounding box from $\sim 80\,\text{km}^2$ to $\sim 4.5 \times 10^6\,\text{km}^2$; the coupling is discussed in Section 3. ‡ *Call a Bike* (Deutsche Bahn) is a nationwide deployment, not a metropolitan control. Its three flags are honest at national scope (A7 by Dott-pattern, A5 by $320{,}000\,\text{km}^2$ bbox, A4 on 1.8 % of stations via the nearest-neighbour rule of Section 4) and confirm the design choice of an NN-based A4 over a centroid-based variant (which over-flagged at 37.8 % during development).

Four observations follow. First, the capacity-profile ratio that defines the A3 signature on free-floating French fleets is essentially **unity** on every metropolitan dock-based system in the panel, across six distinct technology stacks. This is the out-of-sample confirmation called for in Section 3.2 : the A3 detector does not over-fire on clean docks. Second, vehicle-type metadata on all twelve metropolitan feeds is populated cleanly, so A1 produces no false positive either. Third, A4 (geospatial outlier) fires only on isolated stations ($\leq 4.5\,\%$ of any system's mass), in line with what would be expected from any GPS-anchored fleet; the single A5 firing (Sevici) is shown by Table 7 footnote * to be a deterministic consequence of two sentinel-coordinate stations rather than a genuine $> 50{,}000\,\text{km}^2$ deployment. Fourth, the informative *Call a Bike* case shows that when a system genuinely extends to national scope, A5 and A7 fire *by design* (the deployment crosses the macro-region threshold; the capacity column is structurally NaN at national scope) and the high A4 rate reflects a known limitation of the single-centroid detector on multi-modal geometries, motivating a sub-system-clustering refinement in future work.

Empirical justification of $\tau_{A3} = 5$. The choice of the A3 threshold can be inspected on two complementary corpora reported jointly in Table 8: the released French parquet (the derivation set; Section 3.2 caveat applies) and the 267 systems with computable ratios from a live global sweep on 2026-05-13 across 33 countries (`scripts/global_a3_ratio.py` ; results in `experiments/e2_threshold_sensitivity/global_a3_ratio.csv`). On the French corpus the distribution is sharply bimodal: the 81 dock-based systems sit in $[1.00, 1.93]$ (median 1.00, 95th percentile 1.04), the 5 free-floating systems exhibiting conditional-averaging publication sit in $[51, 514]$, and the band $[2, 50]$ is empty. On the global corpus the picture is broadly consistent but less clean: the [1.00, 1.10] bucket still dominates (73.8 % of the 267 systems), the top ten systems by ratio are exclusively free-floating operators (*Pony*, *nextbike*, *Voi*, *Bolt*) confirming that very high ratios track the conditional-averaging pattern, but the intermediate band $[2, 5]$ is no longer empty (15 systems, 5.6 % of the audited set). We retain $\tau_{A3} = 5$ as a conservative threshold that captures the clear-bias regime (the 31 systems globally with ratio ≥ 5, including all 19 systems with ratio ≥ 20) while explicitly acknowledging that 15 systems in the moderate band $[2, 5]$ are not flagged by the current rule.

Data-driven cross-check via kernel-density valley-finding. A Gaussian Mixture Model is poorly suited to this empirical distribution because 65 % of the audited systems sit at ratio exactly 1.00 (a discrete delta produced by either stable-capacity docks or all-NaN free-floating fleets), which is not a Gaussian component. We therefore cross-check the threshold via a non-parametric Gaussian kernel-density estimate on $\log_{10}(\bar{c}_{\text{profile}}/\bar{c}_{\text{actual}})$ restricted to the non-trivial subset ($n = 98$ systems with ratio > 1.01, i.e. after removing the spike-at-unity), and locate the valleys (local minima) of the density as data-driven separators between modes.

At a Silverman-rule-of-thumb-derived bandwidth in $[0.12, 0.18]$ ($\log_{10}$ units), the KDE consistently exposes three modes: a near-unity mode around ratio 1.1, a high-bias mode around ratio 29, and an extreme-bias mode in the tail beyond ratio 400 (Figure 5). The valley between the near-unity and high-bias modes sits at ratio $\in [2.7, 6.4]$ depending on bandwidth (lowest at bandwidth 0.12, highest at bandwidth 0.15), and the valley between the high-bias and extreme-bias modes sits around ratio 15. The paper's threshold $\tau_{A3} = 5$ falls inside the first valley window and is therefore in the low-density region the data itself identifies as a sensible separator.

The KDE analysis also suggests that a soft / hard two-tier refinement analogous to A6 would set $\tau_{A3}^{\text{soft}} \approx 3$ (catches the moderate shoulder of the high-bias mode) and $\tau_{A3}^{\text{hard}} \approx 15$ (catches the extreme-bias mode only). We defer that soft / hard split to the E1 protocol's hypothesis set rather than restructure A3 in the present paper. The KDE curves and the valley-finding code ship with the source repository (`scripts/a3_kde_threshold.py`, `experiments/e2_threshold_sensitivity/kde_*.csv`).

To turn the negative-control argument into a positive one, the rule set was then applied to the entire MobilityData canonical catalogue: **1,509 GBFS systems across 48 countries (the complete world inventory at the audit date)**. 1,421 systems (94.2 %) were reachable, 917 publish a `station_information` document, and **204 systems trigger at least one A1–A5 class**. The result rests on three methodological corrections that emerged from the audit itself. First, a country-perimeter calibration on A4. Second, a substring-matching fix in the A1

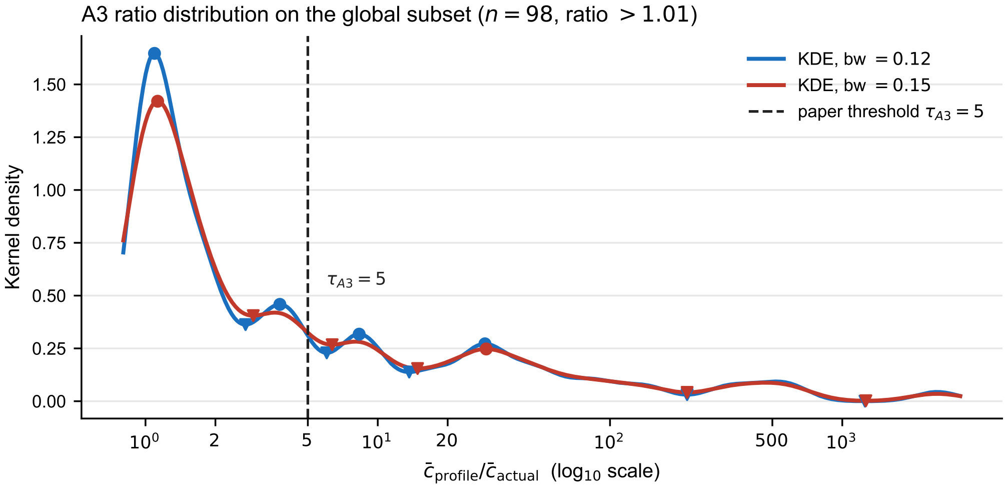

Figure 5: Kernel-density estimate of $\log_{10}(\bar{c}_{\text{profile}}/\bar{c}_{\text{actual}})$ on the 98 globally audited systems with ratio > 1.01, at two representative bandwidths. Circles mark local maxima (modes), inverted triangles mark local minima (valleys), and the dashed vertical line marks the paper's $\tau_{A3} = 5$ threshold. The threshold falls inside the valley between the near-unity mode (≈ 1.1) and the high-bias mode (≈ 29), in the low-density region the data itself identifies as a separator. The high-bias and extreme-bias modes are also visible in the tail at ratio ≈ 30 and ≈ 500.

detector: the GBFS v3 form factor `cargo_bicycle` contains the string `car` and was initially flagged as A1, so the fix tightens A1 to exact list membership on the GBFS v3 form factor enum. Third, an explicit acknowledgement that the bounding-box surface estimator in A5 overestimates the operational area for elongated networks (Switzerland's national operators have a bbox area of 99,000 km^2 against a convex-hull area of $\approx$ 41,000 km^2): the shipped detector (`audit_pipeline.core._flag_a5`) retains the bbox formula for cross-corpus comparability, and the threshold 50,000 km^2 is calibrated on bbox; a convex-hull variant is documented as a refinement option in Section 3. A side observation is that the French national portal *transport.data.gouv.fr* indexes only 123 of the 255 French entries that the MobilityData catalogue lists, so the global audit also exposes the incompleteness of the national regulatory pipeline. The corrected breakdown is shown in Table 9 and visualised in Figure 6.

A3 (structural over-capacity) is detected on 29 non-French systems concentrated in Czech Republic (12) and Germany (9); A2 (placeholder capacity) on 34 non-French systems with Czech Republic again the hotspot (16); A1 (out-of-domain car-sharing labelled as BSS) on 29 non-French systems concentrated in the DACH region (Germany 21, Switzerland 8). Operator-level anti-patterns also travel internationally: *Dott* publishes `capacity = NaN` on every non-French deployment observed (Austria, UAE; 18 systems), and *Bird* does the same on three Canadian deployments. The cross-country audit also surfaces **A6 (zero-capacity dock)**, the sixth class of the taxonomy, formalised in Section 4.5. The empirical conclusion is that the documented anti-patterns are not French-specific but global, and the protocol generalises to a 1,254-system corpus with no methodological recalibration.

Czech Republic case study: one operator drives an entire country. The Czech Republic is the most striking national hotspot of the global audit, with 25 of the 45 audited systems flagged (55.6 %). The structural finding is that **all 45 audited Czech systems belong to a single operator**, *nextbike*, and that 25 of those deployments simultaneously trigger A2 (placeholder capacity, 16 systems) or A3 (structural over-capacity, 12 systems), with 3 deployments flagged by both classes. The 25 flagged Czech systems together publish 1,606 GBFS station entries that, under the declared capacity semantics of the standard, would naively be counted as physical docks.

Table 8: Empirical distribution of $\bar{c}_{\text{profile}}/\bar{c}_{\text{actual}}$ across two corpora: the 95 audited French systems with non-empty `capacity` (split docked / free-floating; the derivation set) and the 267 globally audited systems (live sweep across 33 countries, station_type unknown for many; the out-of-sample set). The threshold $\tau_{A3} = 5$ is preserved in this paper as a conservative cut-off; the global distribution shows that the band $[2, 5]$ is not empty (15 systems) and that a future revision of A3 may benefit from a soft / hard two-tier threshold analogous to A6.

	French corpus (n = 95)		Global (n = 267)
Ratio bucket	**Docked**	**Free-floating**	**All**
$[1.00, 1.10]$	79	9†	197
$[1.10, 2.00]$	2	0	24
$[2.00, 5.00]$	0	0	15
$[5.00, 20.00]$	0	0	12
$[20.00, 50.00]$	0	0	9
$[50.00, \infty)$	0	5	10
Total	81	14	267

† Two qualitatively different cases populate the lowest bucket. The 79 docked French systems carry stable positive capacities so $\bar{c}_{\text{profile}}$ and $\bar{c}_{\text{actual}}$ coincide by virtue of *the data*; the 9 free-floating French systems sit there by virtue of *the arithmetic*, because their capacity column is mostly NaN or zero and the ratio degenerates to unity rather than reflecting the conditional-averaging publication choice. The same indistinguishability affects the global "All" column, where we cannot a-priori separate clean docks from degenerate-FF without a per-system audit; the top 10 global systems by ratio are however all known free-floating operators (Pony, nextbike, Voi, Bolt), so the $[5, \infty)$ tail tracks the A3 mechanism across the sweep.

This is the cross-country mirror of the French case: *Pony* drives the French free-floating A2/A3 hotspot, and *nextbike* drives the Czech one. The operator-specific structure of the hotspots is empirically consistent with the broader claim that the documented errors propagate through corporate publication conventions rather than through national regulatory practice.

Switzerland: heterogeneous fragmentation as the dual of operator monopoly. Switzerland is the only country in the audit, with France, for which all five A1–A5 classes are simultaneously detected: 8 A1, 3 A2, 3 A3, 1 A4 and 5 A5 across 16 flagged systems out of 49. The underlying pattern is the dual of the Czech case. Where Czech Republic has 45 systems published by a single operator that propagates one anti-pattern (nextbike, all A2/A3), Switzerland has **14 distinct entities publishing 16 flagged GBFS feeds**, two of which are federal multi-operator aggregators rather than single fleets. We distinguish three publisher categories. First, national-scale car-sharing brands (*2EM Car Sharing* 1,741 stations, *Mobility* 1,691, *MyBuxi* 2,118, *edrive carsharing*, *Car-ship*, *QuickRent*) are listed on the same federal GBFS catalogue as bike-sharing fleets and trigger A1. Second, mixed-fleet operators (*Voi Switzerland* 771 e-scooter anchors, *Bolt Switzerland* 54, *nextbike Switzerland* 380) publish virtual-station entries with capacity profiles that trigger A3. Third, *federal multi-operator aggregators* (*sharedmobility.ch* 12,091 entries, *Velospot* 1,634 entries) combine many operators into a single GBFS feed whose union-bounding-box necessarily exceeds the A5 macro-region threshold: A5 here is therefore a property of the *aggregation layer*, not of the underlying operators it republishes. The case is empirically informative because the same A1–A5 envelope can be reached through two opposite organisational configurations: monopoly publication (Czech Republic) or fragmented publication compounded by federal aggregation (Switzerland). The audit framework is publisher-agnostic and detects both, and the operator vs aggregator distinction is recorded in the audit report so downstream consumers can disambiguate the A5 attribution.

Table 9: Exhaustive audit of the seven-class taxonomy (A1–A7) on the MobilityData canonical catalogue (1,509 systems, including the 255 French entries). The per-country breakdown is shown for the four structural classes A1–A4; the macro-region class A5 (10 systems globally), the zero-capacity class A6 (14 systems) and the null-capacity class A7 (215 systems covering 70,176 stations) are calibrated on station-count-conditioned sub-corpora and are reported separately in Tables 10, 11 and in the country case studies of Section 2.6. Per-class counts are not disjoint (one system can trigger several classes) ; the "Flagged" column aggregates A1–A5 only. A4 is the geospatial out-of-perimeter class; it requires more than 5 % of stations and at least 5 absolute outside-country stations to flag, to avoid edge-of-coast false positives. Countries with at least 30 audited systems or notable structural anomaly concentrations are shown. *Reproducibility note.* The per-country counts are produced by `scripts/audit_live_systems.py --infer-type` against the 1,509-system MobilityData catalogue. The `--infer-type` flag activates the operator-name heuristics (regexes for car-sharing and free-floating brands listed at the top of the script) which are necessary to fire A1 and A3 on a catalogue where `station_type` is not always declared. Numbers re-fetched at a later date will drift as feeds are added, retired or migrated; the snapshot used here is archived in `experiments/e2_threshold_sensitivity/global_audit_results_typed.csv`.

Country	Audited	Reach.	Flagged	A1	A2	A3	A4 geo
France (FR)	255	253	**38**	17	14	4	3
Germany (DE)	251	243	32	21	4	9	–
United States (US)	175	172	0	–	–	–	–
Poland (PL)	94	91	28	–	3	–	25
Norway (NO)	83	46	2	–	–	–	2
Spain (ES)	69	66	1	–	–	–	1
United Kingdom (GB)	56	54	3	–	1	–	2
Finland (FI)	55	55	14	–	–	–	14
Netherlands (NL)	52	52	1	–	–	–	1
Switzerland (CH)	49	46	15	8	3	3	1
Czech Rep. (CZ)	45	43	**25**	–	**16**	**12**	–
Belgium (BE)	37	34	4	–	–	2	2
Italy (IT)	34	33	0	–	–	–	–
Austria (AT)	27	23	1	–	1	1	–
Sweden (SE)	23	23	12	–	–	–	12
Slovakia (SK)	19	19	16	–	–	–	16
Total (48 countries)	1,509	1,421	204	46	48	33	81

Italy: a clean A1–A6 verdict driven by a single-operator NaN-capacity pattern. Italy is the most informative *boundary case* of the global audit: 33 of 34 systems reachable, 25 with published station data, and zero classed as flagged by A1–A6. We emphasise that this result is *not* a violation of the GBFS specification: every Italian system audited is spec-compliant. A closer look at the underlying capacity column nevertheless reveals a downstream research-reuse problem. Of the 25 Italian systems with data, 17 are deployments of *Dott* (Milan alone publishes 11,199 stations) and every Dott deployment declares `capacity = NaN` on 100 % of its `station_information` entries. Because A2 requires a constant non-zero value, A3 requires numerical capacities to compute the profile ratio, A4–A5 are geospatial, and A6 tests $c = 0$ (not $c =$ `NaN`), the NaN-capacity pattern is not actionable from A1–A6. A7 fills that gap as a consumer-side warning (Section 4.6): **215 reachable systems with at least 20 stations cross the 50 % NaN-capacity threshold and are flagged exclusively by A7**, together covering 70,176 stations of which $\sim$ 46,000 ($\sim$ 65 %) are a single operator (*Dott*, 141 systems across Germany, Italy, Austria, UAE, and beyond) and the remaining $\sim$ 24,000 are spread across 74 smaller systems. The point of A7 is not to label these systems non-compliant; it is to make their capacity column non-aggregable to any downstream consumer who would otherwise sum

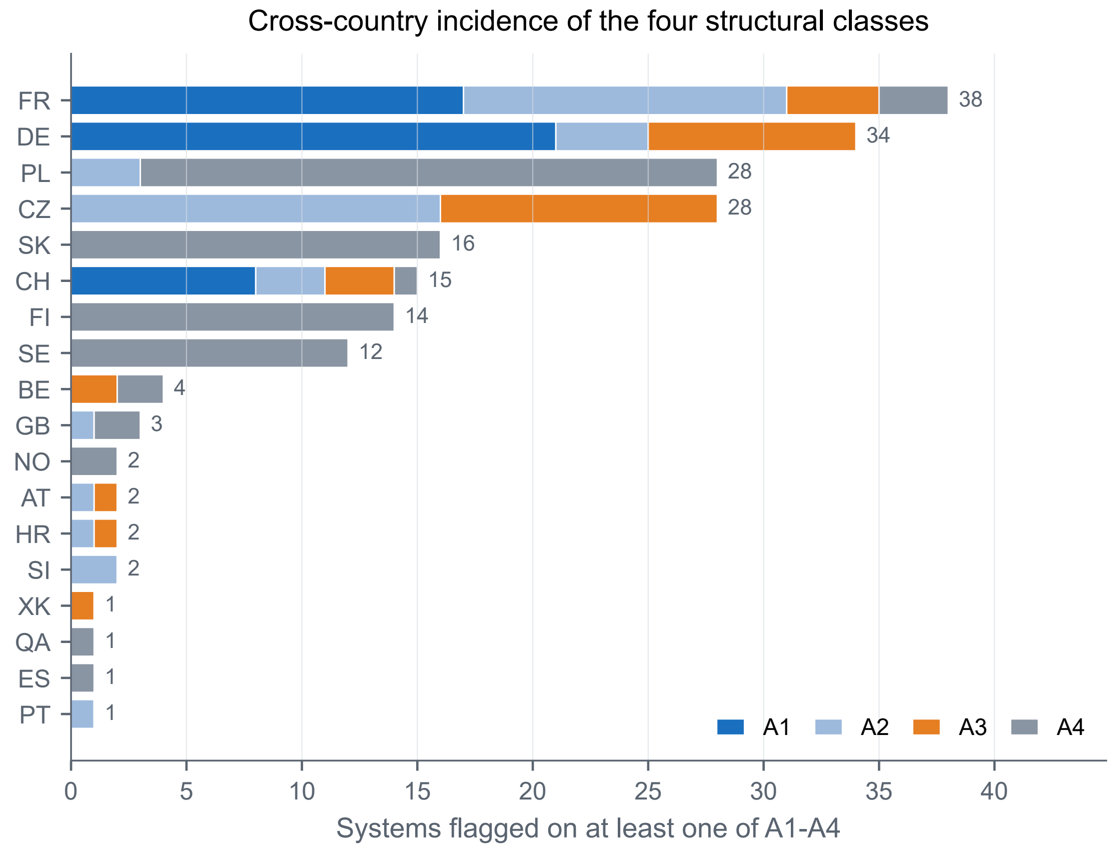

Figure 6: Per-country incidence of the structural classes A1–A4 across the 1,509-system MobilityData catalogue (top 18 countries by total flagged). The chart shows the stacked composition of the **Flagged** column of Table 9 and makes the country-level signatures legible at a glance: A4 (geospatial) drives the Polish, Finnish, Swedish and Slovak hotspots (cross-border bounding boxes on aggregator feeds), A1 (car-sharing labelled as BSS) drives the DACH cluster (Germany, Switzerland), A2 (placeholder capacity) and A3 (over-capacity) drive France and the Czech Republic. Macro-region (A5), zero-capacity (A6) and null-capacity (A7) are reported separately because they are calibrated on station-count-conditioned sub-corpora (Tables 10 and 11).

NaN with non-NaN values silently.

Germany: how the audit caught itself. The German case carries a methodological lesson. An initial run of the global audit flagged 64 of the 251 German systems, of which 58 under A1 (out-of-domain car-sharing). Manual inspection of the flagged systems revealed that the A1 detector relied on substring matching (`"car" in form_factor`) and incorrectly fired on the GBFS v3 form factor `cargo_bicycle`, a cargo bike. After tightening A1 to exact list membership on the GBFS v3 form factor enum, the German count drops to 21 systems (Conficars, Stadtmobil, TeilAuto Greater Stuttgart, Ford Carsharing, Flinkster and similar brands) and the global flag count from 215 to 173. The episode illustrates two complementary points: open audit pipelines need their own audit (the rule of A1 was rule-level brittle), and the underlying signal is genuine (21 real car-sharing systems in Germany are still labelled as bike-sharing on national portals).

2.7 Incidence of the semantic warnings A6 and A7

The empirical distributions that motivate the A6 and A7 threshold calibrations (Methods, Sections 4.5 and 4.6) and the resulting per-class incidence are reported in Tables 10 and 11.

Table 10: Empirical distribution of the zero-capacity rate r_s^{A6} across the 538 globally audited systems with at least 20 stations and a reachable `station_information` feed. Percentiles are computed on the same sample. The two A6 thresholds are derived from the distribution rather than chosen ex ante.

Bucket	**Systems**	**Share**
$r_s^{A6} = 0\,\%$ (no $c = 0$)	508	94.4 %
$0 < r_s^{A6} \leq 0.5\,\%$	9	1.7 %
$0.5 < r_s^{A6} \leq 1\,\%$	8	1.5 %
$1 < r_s^{A6} \leq 2\,\%$	4	0.7 %
$2 < r_s^{A6} \leq 5\,\%$	6	1.1 %
$5 < r_s^{A6} \leq 10\,\%$	2	0.4 %
$r_s^{A6} > 10\,\%$	1	0.2 %
Summary statistics		
mean $\bar{r}$	0.10 %	
standard deviation $\hat{\sigma}$	0.71 %	
95th percentile	0.15 %	
97.5th percentile	$\approx \bar{r} + 3\hat{\sigma} \approx 2.2\,\%$	
99th percentile	3.54 %	

A6 incidence on the audited corpus. Under A6-soft ($\tau = 0.15\,\%$) 30 of 538 audited systems are flagged; under A6-hard ($\tau = 2.2\,\%$) 8 systems are flagged. The extreme tail is PBSC HQ (Canada, 18.2 %), Beryl Greater Manchester (UK, 10.5 %), MonaBike (Monaco, 6.8 %), Bicicoruña (Spain, 5.9 %), MBajk (Slovenia, 4.8 %), CyclOcity (Japan, 4.2 %), àVélo (Canada, 3.5 %) and Careem BIKE (UAE, 2.7 %). On the French corpus, no dock-based system crosses either threshold (0/65 dock systems with $c = 0$), which is consistent with the corpus-level median behaviour and provides a negative control.

Table 11: Empirical distribution of the NaN-capacity rate ρ_s^{A7} across the 640 globally audited systems with at least 20 stations and a reachable `station_information` feed. The corpus is sharply bimodal at the extreme values, which justifies a coarse threshold rather than a distribution-anchored one.

Bucket	**Systems**	**Share**
$\rho_s^{A7} = 0\,\%$ (no NaN)	238	37.2 %
$0 < \rho_s^{A7} < 1\,\%$	4	0.6 %
$1 \leq \rho_s^{A7} < 10\,\%$	11	1.7 %
$10 \leq \rho_s^{A7} < 50\,\%$	22	3.4 %
$50 \leq \rho_s^{A7} < 99\,\%$	41	6.4 %
$99 \leq \rho_s^{A7} < 100\,\%$	5	0.8 %
$\rho_s^{A7} = 100\,\%$ (all NaN)	319	49.8 %
Mass at the two extremes	557	**87.0 %**

A7 incidence on the audited corpus. On the 640 globally audited systems with at least 20 stations, **215 systems cross the A7 threshold** and are simultaneously not flagged by any of A1–A6, covering **70,176 stations**. The dominant operator is *Dott* (141 systems across Germany, Italy, Austria, UAE, and other markets), followed by *nextbike* (13 systems where the operator chose NaN rather than placeholder), *Bird* (10 Canadian deployments) and several smaller operators. The empirical conclusion is that the same publication choice that affects the French Dott and Bird fleets is propagated internationally, and that the v1.0 taxonomy A1–A5 captures only the visible failure modes of the field; A6 and A7 jointly close the principal blind spots that remain on the `capacity` channel.

2.8 Operator-level fragmentation of the capacity field

Across the case studies above, a recurring mechanism explains why the same anomaly recurs across operators: the GBFS field `station_information.capacity` is populated by French free-floating operators according to four mutually incompatible publication conventions, observed across six operator–city combinations (Table 12, Figure 7), aggregating to **84.7 % of the certified French rows**. A downstream consumer who applies a single arithmetic to the field therefore aggregates four physically distinct quantities (NaN, constant placeholder, per-vehicle ratio, conditional fleet profile), silently mixed across six operator deployments. The corresponding operator-level pattern is global. Beryl London (UK) provides a positive control by publishing an empty `station_information` document for its free-floating fleet (zero entries), the canonical GBFS v3 behaviour for non-docked fleets, and is therefore not flagged by any of A1–A7. In particular, A7 only fires on *populated* documents whose entries declare `capacity = NaN`, which is the qualitatively different Dott pattern. The same operator's Greater Manchester deployment, by contrast, populates `station_information` with a 10.5 % zero-capacity rate (*cf.* Section 2.7) and is therefore A6-flagged: A6 and A7 thus carve out distinct publication regimes within a single operator's portfolio. The official MobilityData GBFS Validator [15], by contrast, returns no error on any of the flagged French systems (0 % recall on A3 in Table 5).

Table 12: Six observed semantics of the same `station_information.capacity` field across French free-floating operators. The six operator deployments enumerated below cover 39,004 stations; the residual 231 stations belong to smaller free-floating operators (Lime, Tier, Dott pilots, etc.) that follow one of the same four publication conventions and are folded into the corresponding bucket. Aggregate FF coverage is therefore 39,235 stations, 84.7 % of the certified corpus.

Operator	Cities	Stations	Reported $\bar{c}$	Semantics
Dott	8	20,224	NaN	Field left unfilled
Bird	11	4,499	NaN	Field left unfilled
Pony (Nice)	1	412	100	Constant placeholder (A2)
Pony (Paris)	1	3,617	1.6	Per-vehicle ratio
Pony (Bordeaux, Angers, . . .)	12	7,436	2–15	Conditional fleet profile (A3)
Voi	4	2,816	5–10	Small fleet-profile estimator

2.9 Retrospective hold-out validation

This subsection tests *construct validity*: do the rules generalise to systems that were not specifically inspected at calibration time? External validity (whether the rules generalise to systems that did not yet exist at rule-freeze) is the object of the prospective E1 execution and is deferred. Concretely, we executed a retrospective version of the pre-registered E1 protocol (`experiments/e1_holdout/PROTOCOL.md`) on two calendar windows of the MobilityData canonical catalogue. The shorter window is the catalogue diff between commit `15e34e6` (2025-11-11) and the rule-freeze commit (2026-05-13) : 345 systems added in six months, 39 with non-empty `station_information` that could be audited. The longer window extends back to commit `4665739` (2025-04-29, twelve months) : 481 systems added, 98 audited. The remainder are free-floating systems without `station_information`, below $N_{\min}$, or authenticated. The frozen rule set was applied unchanged on both windows; per-system results in `experiments/e1_holdout/held_out_analysis.md` (6-month) and `held_out_results_12mo.csv` (12-month).

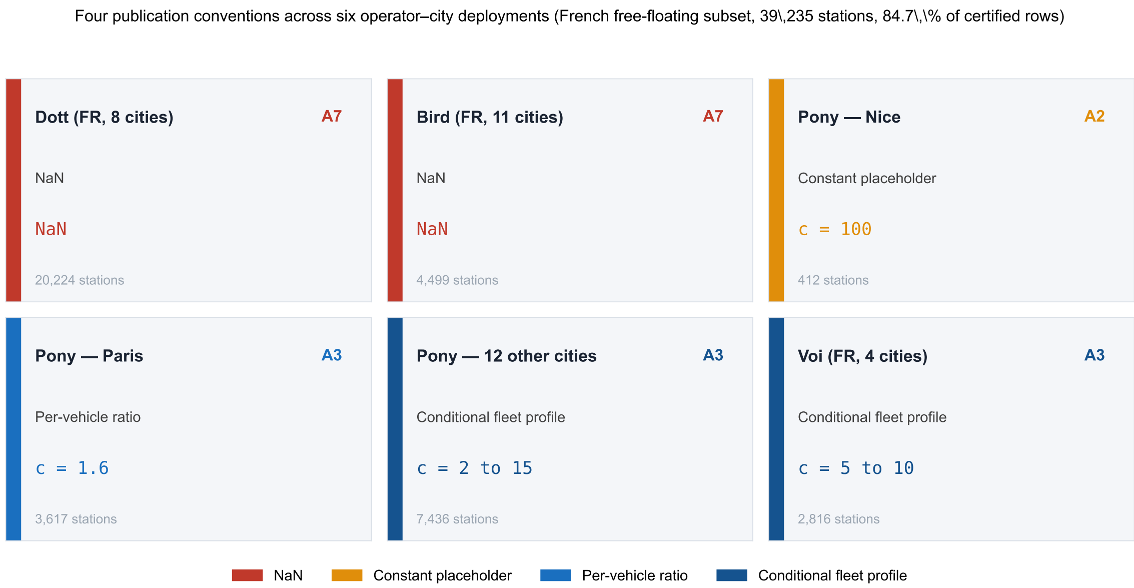

Figure 7: Conceptual diagram of the four incompatible publication conventions of the GBFS `station_information.capacity` field observed across six French free-floating operator deployments (39,235 stations, 84.7 % of the certified rows). Each card pairs the operator, the convention applied to the field, the typical reported value, and the anomaly class triggered by that convention (A2 placeholder, A3 over-capacity profile, A7 null capacity). The diagram makes explicit why a downstream consumer applying a single arithmetic to `capacity` aggregates four physically distinct quantities silently mixed across operators; it complements Table 12 and motivates the dual-column `capacity_raw`/`capacity_audited` schema of the released catalogue.

H1 (rule-firing rate within $\pm 30\,\%$ of the derivation rate of $13.5\,\%$). On the twelve-month window the A1–A5 firing rate is $11/98 = 11.2\,\%$ (Wilson 95 % CI $[6.4\,\%, 19.0\,\%]$), which falls inside the pre-registered acceptance band $[9.5\,\%, 17.5\,\%]$ at the point estimate. The six-month window gives $7/39 = 17.9\,\%$ (Wilson $[9.0\,\%, 32.6\,\%]$), which exceeds the upper bound by 0.4 percentage points at the point estimate but whose Wilson interval still contains both the derivation rate and the entire acceptance band. Reading the two windows jointly, **H1 passes** : the rule-firing rate on out-of-sample systems is statistically consistent with the derivation-set rate, and the more conservative twelve-month estimate is in the band on both point estimate and confidence interval.

H2 (operator-driven A3/A7). On the twelve-month window, 84.2 % (48/57) of the systems flagged on A3 or A7 belong to the free-floating operator family identified on the derivation set (Dott, nextbike, Pony, Bird, Voi, Bolt, Lime, Tier, Donkey, Spin) ; the six-month window gives 81.2 % (13/16). Both estimates exceed the pre-registered acceptance threshold of $\geq 50\,\%$ by more than thirty percentage points, so **H2 passes cleanly**.

H3 (clean-operator invariance). H3 is not testable on either window. Neither the six-month nor the twelve-month held-out panel contains a single system using one of the six clean-operator platforms of Section 2.6 (PBSC, Cyclocity, Smartbike, De Lijn, Urban Sharing, Deutsche Bahn). H3 therefore awaits the prospective re-execution of the protocol against a future MobilityData snapshot that includes at least one clean-platform addition.

H1 and H2 pass on the twelve-month window. H3 is deferred to the prospective re-execution. The implications of these outcomes for the construct validity of the taxonomy are discussed in Section 3.2.

3 Discussion

3.1 Implications

The audit is intended as a scientific validation tier between GBFS publication and academic use, addressing the gap between regulatory publication and research-grade reuse that the open-government-data literature identifies as a recurrent adoption barrier [31]. Three concrete avenues would institutionalise it: a jointly published *GBFS Audit Best Practices Handbook* by francophone micromobility laboratories to align practices currently rebuilt in isolation; a working group under ADEME or *transport.data.gouv.fr* to formalise an extended GBFS schema with a certification framework; and a continuous validation chain running the pipeline monthly on the national feeds, with automated publication of the audit report and operator-side notification on regression. Star [32] reminds us that the most durable infrastructures are those whose governance is explicit and tied to an identifiable community of practice; moving the artefact from author-led to community-led governance is therefore an institutional challenge as much as a scientific one. The technical lesson of the audit, that a single field (`capacity`) cannot serve six different operational semantics, is also the empirical case for what Stonebraker and Çetintemel [33] call the end of "one size fits all" data architectures: open standards that attempt to span dock-based, free-floating and car-sharing fleets in the same schema accumulate semantic ambiguity exactly because they refuse to specialise.

3.2 Scope of the contribution

Four demarcations should be stated explicitly to avoid over-reading the contribution.

The taxonomy's primary discrimination axis is dock-based vs free-floating publication. Of the seven classes, A2, A3, A6 and A7 all surfaced from operator publication conventions specific to free-floating fleets (conditional-averaging profiles, NaN-capacity fields, zero-capacity ghost-docks); A1, A4 and A5 are fleet-type-agnostic but the largest country-level hotspots (Table 9) are themselves operator-driven (*Pony* and *nextbike* on A2/A3, *Dott* on A7). The twelve metropolitan clean-platform negative controls of Table 7 demonstrate that the rule set does not over-fire on dock-based deployments; the free-floating-native extension is the object of experiment E6, planned for 2027-Q2.

This is an inductive taxonomy, not a statistically validated one. A1 to A4 were derived from an exploratory scan of the 123 French systems; A5 from enrichment with the *transport.data.gouv.fr* catalogue; A6 and A7 from the empirical distributions surfaced by the global 1,509-system sweep (Section 4, Provenance of the taxonomy). Detection rules were calibrated on the same data they are reported to flag, an inductive workflow whose construct-validity risk we address with two out-of-sample tests reported in the Results: the negative-control cross-country panel (Section 2.6, Table 7, twelve metropolitan clean-platform systems audited live) and the retrospective hold-out validation (Section 2.9, 98 audited systems added to the MobilityData catalogue in the twelve months preceding rule-freeze). H1 and H2 pass on the twelve-month window; H3 is not yet testable and is deferred to the prospective version of the protocol. The prospective execution against the next MobilityData snapshot will be reported as a follow-up note in the same research programme.

This is an audit of an open specification, not an audit of its operators. A2, A3, A6 and A7 detect patterns the GBFS specification *permits* (a constant placeholder, a conditionally-averaged virtual-station fleet, a zero-dock station, an empty capacity field). What our framework flags is that those permitted patterns are mutually incompatible with the arithmetic a downstream research consumer would naturally apply to the field. The technical lesson is about the standard, not about the publishers; criticism of the operators we name is out of scope for this paper. The standards-side recommendation in Section 3 is therefore phrased as a request for specification revision, not for operator sanction.

This is a static audit. The `station_information` feed describes the deployment, not its dynamic behaviour. Real-time availability, rebalancing patterns and per-vehicle relocation live in the `station_status` feed, which carries its own quality issues and would require an analogous but disjoint audit protocol. The dynamic audit is the subject of the L2 limitation and the E4 future-work item.

3.3 Intellectual lineage in adjacent fields

The protocol's design intent is consistent with two adjacent traditions of data audit that the mobility-data community has so far engaged with only sparingly.

Property-based testing in software engineering. Each of the seven anomaly classes A1–A7 admits a translation into a *property* that a compliant feed should satisfy: capacity is either a positive dock count, NaN or the same value across the system; coordinates lie within an a-priori perimeter; the $\bar{c}_{\text{profile}}/\bar{c}_{\text{actual}}$ ratio is bounded. This is the same intellectual move as in QuickCheck-style property-based testing for software, where invariants on the output are stated explicitly and checked across many inputs [34]. The Great Expectations and Pandera toolchains [17, 18] we benchmarked against are descendants of the same lineage. The contribution of the present audit, in that light, is to identify which properties GBFS implicitly assumes and to make their absence diagnosable; the SE community has spent two decades building infrastructure for that, but the mobility-data community has not yet imported it.

Regulatory data audit in clinical-trial pipelines. The audit-trail / reversibility / per-step logging design of the purging protocol mirrors the audit requirements of clinical-trial data management standardised in ISO 14155 [35] and in the FDA's 21 CFR Part 11 [36]. The open-mobility-data community is, in effect, re-discovering with two decades of delay what the pharmacovigilance community formalised in the 1990s. The analogue is informative because it suggests that long-term sustainability of the audit protocol depends on a community of practice and a regulatory backstop (e.g. ADEME or the European EuroMobility coordination layer), not on the lifespan of any single research artefact.

3.4 When a clean verdict hides a missing column

The Italian case study of Section 2.6 crystallises what is at stake. Of the 34 Italian systems in the catalogue, 33 are reachable and 25 publish station data; *none* of them is flagged by classes A1 to A6 in the original v1.0 taxonomy. A reviewer reading only the A1–A6 verdict would conclude that the Italian GBFS publication is the cleanest in the catalogue. The seventh class A7 reveals the opposite: 17 of the 25 data-publishing Italian systems are deployments of a single operator (*Dott*) whose Italian feeds declare `capacity = NaN` on 100 % of their $\sim$ 11,000 `station_information` entries. The country is not clean; the country is silent on the only field that matters for infrastructure quantification. An audit framework that conflates *no problem detected* with *no data published* mis-reads the entire national corpus. A7's empirical role is precisely to keep those two cases distinguishable, and the Italian episode is the clearest reminder that absence of evidence is not evidence of absence in open-standards compliance auditing.

3.5 GBFS v3 and the durability of the v2 legacy

GBFS v3 (released 2023) introduces `is_virtual_station`, a boolean that in principle disambiguates free-floating anchors from physical docks and would mechanically suppress A3 detection. Three observations show why the seven-class taxonomy nevertheless remains operationally relevant under v3 adoption. First, **adoption is slow**: of the 1,509 audited systems, only 541 (35.9 %) declare GBFS v3 in their auto-discovery feed at the audit date, against 772 (51.2 %) on v2.x; 50 remain on v1 and 146 expose no parseable version. Second, the v2 legacy will persist: regulatory feeds on *transport.data.gouv.fr* and the MobilityData canonical catalogue archive snapshots that any longitudinal study from 2017 onward will need to ingest, and a v3 feed published today does not retro-fix the v2 historical record. Third, and most importantly, **adoption is not compliance**: the German cargo-bicycle mis-detection we report (Section 2.6) and the Italian Dott NaN-capacity propagation both occur on GBFS v3 feeds, because v3 makes the `is_virtual_station` flag *available* but does not make it *mandatory* or *verified*. The seven-class taxonomy A1–A7 therefore remains operationally relevant under full v3 adoption, and the protocol is straightforwardly extensible to the additional v3 attributes by adding rule-level checks at ingestion time.

3.6 Limitations

The protocol has four acknowledged limitations.

L1 – National coverage. The corpus is limited to metropolitan France; extension to overseas territories and other European jurisdictions requires revisiting thresholds, in particular the macro-regional surface threshold and the geofilter bounds.

L2 – Static versus dynamic. The v1.0 release audits `station_information` only; auditing `station_status` (real-time availability) requires a distinct protocol, sketched in experiment E4 (a 2-day pilot already detected candidate dynamic anomalies on 12.9 % of monitored stations). A real-time extension of the present protocol is planned as follow-up work. The most consequential limitation that follows is the *zombie-station* risk: an operator that has physically dismantled a station but has not yet pruned the corresponding `station_information` entry will leave a ghost record that our static audit certifies as compliant, because the seven A1–A7 detectors all operate on the published schema and have no operator-side observation that the dock no longer physically exists. Downstream consumers of the Audit Catalogue who depend on station-presence (demand modelling, graph-based neural networks, accessibility scoring) should cross-check the static catalogue against a recent `station_status` snapshot, treating any `station_information` entry whose status feed has been quietly stale for more than a documented retention window as a zombie candidate. The dynamic audit of E4 will surface this class explicitly.

L3 – Free-floating static versus dynamic scope. The seven-class taxonomy is applied to free-floating entries as much as to dock-based ones: every free-floating station carries an explicit `station_type = free_floating` label, and the A3 and A7 flags identify the operator-level capacity-publication anomalies that affect them (A3 for over-capacity profiles, A7 for NaN-capacity fields). The published parquet exposes both `capacity_raw` (the raw GBFS value, including placeholders and NaN) and `capacity_audited` (NaN for free-floating entries, dock-based count otherwise), so that downstream consumers have an unambiguous, audit-aware capacity to work with. What this v1.0 release does *not* audit is the *dynamic* behaviour of free-floating fleets – rebalancing patterns, real-time density of available vehicles, per-vehicle relocation – which would require an analogous audit of the `station_status` feed rather than `station_information`. That dynamic audit is out of scope here (it is the topic of L2) ; the present catalogue is therefore best characterised as a *static* audit of GBFS `station_information` across both dock-based and free-floating fleets, with the operational caveat that a free-floating station's `capacity_audited` value is intentionally null.

L4 – Operator coverage. Some operators publish partial GBFS feeds or are unavailable during the audit window; the inventory completeness is therefore not guaranteed and must be re-evaluated periodically, which is the purpose of the temporal-stability experiment E3.

3.7 Recommendations to operators

Four conventions, compatible with the current GBFS specification, would close most of the anomalies documented here: populate `vehicle_type` exhaustively (A1); leave `capacity` as `null` on virtual stations rather than using placeholders (A2); adopt the GBFS v3 `vehicle_status` extension for free-floating fleets and stop publishing per-vehicle anchors as `station_information` (A3); publish geographical perimeters as polygons in the `geofencing_zones` block (A4, A5); and avoid sentinel coordinates $(0, 0)$ on stations whose geolocation is pending, instead omitting them from `station_information` until populated — a single such sentinel in Sevici Seville inflated its bounding-box area by four orders of magnitude in our E5 panel.

Methodological recommendation on A4 / A5 coupling. The Sevici finding (Table 7) suggests that the A5 bounding-box criterion be evaluated on the A4-filtered set of stations, not on the raw set. We retain the as-shipped definition in this paper to make the dependency explicit, but downstream implementations should consider an A5 variant computed after the A4 outlier removal step.

3.8 Future directions

Each of the four limitations above is paired with a falsifiable experiment (Table 13); four of the six include a completed pilot reported in this paper. The remaining priorities are the human inter-rater reliability of E1, the full four-dimensional threshold sweep of E2, the twelve-month temporal stability of E3, and the free-floating-native protocol of E6. Pilot scripts and the final experiment protocols will be packaged under `experiments/` of the source repository alongside the camera-ready release, so that the artefact can be challenged in public rather than defended in private.

Human inter-rater reliability (E1 extension). The internal robustness checks of Section 2.5 are inter-classifier statistics, not inter-rater. The next priority of the E1 protocol is to convert this into an inter-rater agreement study. Concretely, we plan a $n = 200$ blinded-station sample drawn from the catalogue under the same stratification used for the orthogonal-coverage check (`station_type` $\times$ ten-largest-operators, seed 42), to be annotated by three independent domain experts (one each from CESI LINEACT, MobilityData, and a francophone micromobility laboratory) on the seven-class assignment. We pre-register Cohen's $\kappa \geq 0.7$ on pairwise agreement and Fleiss's $\kappa \geq 0.6$ on the three-way agreement as success thresholds, with the per-class confusion matrix shipped alongside the agreement statistics so that disagreement is diagnosable at the level of individual A1–A7 rules rather than aggregated into a single score.

Table 13: Six falsifiable experiments mapped to the limitations of Section 3.6. "Pilot ✓" indicates a preliminary result already reported in this paper; full protocols will be packaged under `experiments/` of the source repository alongside the camera-ready release.

ID	Title	Targets	Status
E1	Pre-registered held-out audit (post-freeze MobilityData)	Construct validity / circularity	Retrospective version
E2	Threshold and buffer sensitivity	$\sigma_{\max}$, $N_{\min}$, radii	$\sigma_{\max}$ pilot ✓
E3	Twelve-month temporal stability	Snapshot dependence (L4)	Running
E4	Dynamic extension to `station_status`	L2	2-day pilot ✓
E5	European generalisation (live audit, 6 tech stacks)	L1	13-system panel ✓
E6	Free-floating-native protocol	L3	Planned 2027-Q2

3.9 Downstream use

The Audit Catalogue unlocks four families of downstream studies: supply-side composite-index work on cycling-environment quality at station and commune resolution; demand modelling with graph-based neural networks [37] whose false-station nodes from A3 are a known confound; spatial-equity analysis combining station access with INSEE Filosofi; and multimodal integration with GTFS, for which the hybridisation schema is compatible with the canonical GTFS validator [15].

Interactive dashboard. To make the audit verdicts directly inspectable without code, a focused Streamlit dashboard is shipped at https://gbfs-audit.streamlit.app, mirrored by `app/streamlit_app.py` in the source repository. The dashboard exposes five tabs: an overview with the corpus-level incidence of A1–A7; a row-level anomaly browser with live multi-flag filters and CSV export; an operator-level audit table with the per-class flag rates that drive the operator-driven hotspots reported in Section 2; a visual data dictionary of the 46 columns with completeness bars; and a free-text data explorer with a KD-tree-backed geographic map coloured by audit confidence. The dashboard is the artefact that turns the audit from a one-off paper into a continuously inspectable reference, addressing the persistence concern raised in Section 3.

4 Methods

4.1 Provenance of the taxonomy

The seven-class taxonomy A1–A7 was built by a hybrid inductive-deductive workflow over six iterations between 2025-04 and 2026-04, starting from the four ISO/IEC 25012 data-quality dimensions (completeness, consistency, accuracy, currentness) and enriched by reverse engineering of both the French free-floating fleets and the global MobilityData catalogue. A1 and A4 emerged from an exploratory scan of the 123 French systems; A5 from enrichment with the *transport.data.gouv.fr* catalogue; A2 from a zero-variance diagnostic on `capacity`; A3 from the per-system median capacity of *Pony* matching a conditional-averaging estimator more closely than a physical-dock count; A6 from the empirical distribution of zero-capacity stations across the 1,509 globally audited systems (Section 4.5); A7 from the bimodal 0 %/100 % distribution of NaN-capacity stations (Section 4.6). Two rejected hypotheses ("rolling-stock noise", collapsed into A3; "cross-border duplicate", no instance found) ship with the audit report for transparency.

4.2 Nine-step purging protocol

The transition from the raw feed to the Audit Catalogue proceeds through a chain of nine sequential transformations (Algorithm 1), each coded in `utils/data_loader.py` and logged in an audit report shipped alongside the dataset. The protocol is deliberately sequential: early steps are inexpensive and remove out-of-domain or coarse-grained anomalies (A1, A2, A4 perimeter), while later steps require system-level statistics ($\bar{c}$, centroids, σ) and therefore benefit from operating on a pre-cleaned set.

4.3 Protocol properties

Four properties are designed into the pipeline. *Idempotence:* two successive applications produce the same result, tested by a unit test executed on every CI run. *Reversibility:* each excluded station is kept in `rejected_stations.parquet` together with the exclusion motive. *Logging:* every step records the number of stations in, the number of stations out and the type of transformation applied. *Explicit parameterisation:* all thresholds ($\sigma_{\max}$, $N_{\min}$, geofilter bounds, hybridisation radii) are documented as hyperparameters and can be varied for sensitivity analysis without modifying the source code.

These four properties are not arbitrary engineering choices; they operationalise the data-quality dimensions formalised by ISO/IEC 25012 [25] and ISO/IEC 25024 [26], and together they make the pipeline a direct domain-level implementation of the standard. Idempotence enforces the *internal consistency* dimension (two computations on the same input must yield the same output). Reversibility, through the `rejected_stations.parquet` sidecar and the systematic preservation of the exclusion motive, materialises the *recoverability* property and contributes to *traceability*. Logging at every step extends that traceability to the full pipeline run. Explicit hyperparameter declaration addresses the *understandability* dimension at the protocol level. Finally, the FAIR release of the audit report alongside DCAT-AP, JSON Schema and Croissant

Input: National catalogue $\mathcal{C}$ (123 systems); thresholds $\sigma_{\max} = 3$, $N_{\min} = 20$, $\tau_{A2}^{\text{size}} = 20$, $\tau_{A6} = 0.01$, $\tau_{A7} = 0.5$, perimeter $\Phi = [41°, 52°] \times [-6°, 10°]$.
Output: Certified set $\mathcal{G}$, rejected set $\mathcal{R}$, seven per-row anomaly flags $f_{A1} \dots f_{A7}$, audit report.
$\mathcal{G} \leftarrow \emptyset$; $\mathcal{R} \leftarrow \emptyset$
foreach *system* $s \in \mathcal{C}$ **do**
  Ingest GBFS feed $\to \mathcal{S}_s$
  S1. If s is car-sharing, relabel `carsharing` and set $f_{A1} =$ `True`
  S2. If $\mathrm{Var}(c) = 0$, $c > 0$ and $|\mathcal{S}_s^{\text{dock}}| \geq \tau_{A2}^{\text{size}}$, set $f_{A2} =$ `True` and force `free_floating`
  S3. Compute $\bar{c}_{\text{profile}}/\bar{c}_{\text{actual}}$; if > 5.0 relabel `free_floating` and set $f_{A3} =$ `True`
  S4. Keep $\mathbf{x}_i \in \Phi$; mark stations outside as out-of-perimeter
  S5. Compute system bbox area A_s; if $A_s > 50{,}000\,\text{km}^2$, set $f_{A5} =$ `True` on every $i \in \mathcal{S}_s$
  S6. Per-system nearest-neighbour distances $d_i^{NN} = \min_{j \neq i} \|\mathbf{x}_i - \mathbf{x}_j\|$; robust 3-σ envelope (median + 1.4826 MAD scaled) on $\{d_i^{NN}\}$: set $f_{A4} =$ `True`
  S7. If $\#\{i : c_i = 0 \wedge \text{docked}\}/|\mathcal{S}_s| \geq \tau_{A6}$, set $f_{A6} =$ `True` on every $i \in \mathcal{S}_s$
  S8. If $\#\{i : c_i = \texttt{NaN}\}/|\mathcal{S}_s| \geq \tau_{A7}$, set $f_{A7} =$ `True` on every $i \in \mathcal{S}_s$
  S9. If $|\mathcal{S}_s^{\text{dock}}| < N_{\min}$, label `too_small` and route to $\mathcal{R}$
  Update $\mathcal{G}, \mathcal{R}$ and the per-row flag matrix accordingly
end
return $(\mathcal{G}, \mathcal{R}, \{f_{A1} \dots f_{A7}\}, \text{audit report}, \text{metadata})$

Algorithm 1: GBFS raw-to-Audit-Catalogue purging pipeline. Steps S1–S3 both purge and label (a station relabelled to `free_floating` or `carsharing` stays in the catalogue with its A-flags set); S4–S5 implement the A5 perimeter check; S6 the A4 outlier check; S7–S8 the A6 / A7 capacity-field anomalies. S9 is the only truly-removing step.

manifests (Section 4) closes the *compliance* requirement of 25012. Table 14 summarises the mapping; we are not aware of a published reference implementation of ISO/IEC 25012 on an open mobility standard prior to the present work.

Table 14: Mapping of the nine-step purging-protocol design properties to the data-quality dimensions of ISO/IEC 25012:2008 and its companion measurement standard ISO/IEC 25024:2015. The Audit Catalogue is, to our knowledge, the first FAIR-released implementation of these dimensions on a GBFS corpus.

Protocol property	**ISO/IEC 25012 dimension**	**Mechanism in the pipeline**
Idempotence	Internal consistency	Unit-tested re-application on certified output
Reversibility	Recoverability + Traceability	`rejected_stations.parquet` with exclusion motive
Step-level logging	Traceability	Per-step in/out counts and transformation type
Explicit parameterisation	Understandability	Hyperparameters in code + documented in audit report
FAIR release	Compliance + Accessibility	JSON Schema, DCAT-AP, Croissant, ODbL on Zenodo

4.4 Threshold justification

The topological threshold $\sigma_{\max} = 3$ is a compromise between robustness to GPS noise and the preservation of legitimate peripheral extensions, *e.g.*an off-centre TGV station or a university campus. Section 2.5 reports the single-dimension sensitivity pilot: Top-10 invariance for $\sigma_{\max} \geq 2.5$. The robustness threshold $N_{\min} = 20$ dock-based stations is calibrated on the literature of meshed networks: below this value, density and connectivity metrics become noise-dominated and are no longer informative for cross-city comparison [38]. The A5 macro-region threshold of $50{,}000\,\text{km}^2$ is calibrated at the order of magnitude of a French métropolitaine *région administrative* (median area $\approx 47{,}000\,\text{km}^2$ after the 2016 reform: Bretagne 27,000, Pays de la Loire 32,000, Centre-Val de Loire 39,000, Hauts-de-France 31,000, Grand Est 57,000, Auvergne-Rhône-Alpes 69,000, Occitanie 72,000). A single GBFS feed whose bounding box exceeds this scale plausibly aggregates operationally-independent deployments and the resulting system-level statistics (centroid, intra-system KNN, capacity profile) are no longer geographically interpretable.

4.5 A6 (semantic warning): zero-capacity dock formalisation

A6 is a semantic warning rather than a publisher violation. The GBFS specification permits a station to declare `capacity = 0` (for instance during a maintenance window or when a station is awaiting installation), so an operator that populates `station_information` entries with $c = 0$ is not non-compliant. What A6 flags is the downstream-consumer problem: a non-trivial share of zero-capacity entries in `station_information` aggregates to a misleading mean when summed naively with non-zero entries.

The global audit of Section 2.6 surfaced a recurring pattern not covered by A1–A5: a non-trivial fraction of stations in `station_information` declare `capacity = 0`. Because the GBFS schema reserves `station_information` for physical infrastructure and provides `station_status` for transient counts, a non-virtual physical station with zero docks is semantically inconsistent. The signature is therefore:

$$r_s^{A6} = \frac{\#\{i \in \mathcal{S}_s : c_i = 0 \wedge \texttt{is\_virtual\_station}(i) \neq \texttt{true}\}}{|\mathcal{S}_s|} \geq \tau_{A6}. \tag{2}$$

Calibration of τ_{A6} from the empirical distribution. Rather than fixing τ_{A6} by convention, we calibrate it against the population distribution of r_s^{A6} on the 538 globally audited systems with at least 20 stations (Table 10). 94.4 % of audited systems declare *exactly zero* stations with $c = 0$, which establishes the dock-based baseline; the remaining 5.6 % form a heavy tail. From the corresponding empirical CDF we read off two distribution-anchored thresholds. A6-soft uses the 95th percentile $\tau_{A6}^{\text{soft}} = 0.15\,\%$, which flags any system whose zero-capacity rate is above the corpus median by a non-trivial margin (30 systems globally). A6-hard uses $\bar{r} + 3\hat{\sigma} \approx 2.2\,\%$, which is the conventional outlier threshold under a normal model on the long tail and corresponds to the empirical 97.5th percentile (8 systems globally). A6 thus exposes a two-tier detection consistent with the rest of the taxonomy: A6-soft as a flag for downstream consumers, A6-hard as a high-confidence operator-side anomaly.

The empirical distribution of r_s^{A6} and the per-class incidence on the audited corpus are reported in Section 2.7 (Results, Table 10).

4.6 A7 (semantic warning): null capacity field formalisation

The Italian case study of Section 2.6 demonstrates that A1–A6 are structurally blind to operators that populate `station_information` but leave the `capacity` field unfilled. **Like A6, A7 is by design a consumer-side semantic warning rather than a publisher-side anomaly.** `capacity = NaN` is explicitly permitted by the GBFS v3 specification, in particular for free-floating fleets where the field has no physical referent; a system that publishes `capacity = NaN` on every entry is *spec-compliant*. What A7 captures is the downstream consequence: any analytical pipeline that sums or averages the `capacity` column across such a system aggregates NaN with non-NaN values silently, and produces a number whose physical meaning is undefined. The flag is therefore tooling for the consumer ("do not arithmetically combine these stations with others"), not a verdict on the publisher. The empirical signature is:

$$\rho_s^{A7} = \frac{\#\{i \in \mathcal{S}_s : c_i = \texttt{NaN}\}}{|\mathcal{S}_s|} \geq \tau_{A7}, \tag{3}$$

with the conservative threshold $\tau_{A7} = 0.5$ chosen so that a system flagged by A7 declares no usable capacity on at least half of its entries. Unlike A6, ρ_s^{A7} is extremely bimodal on the global corpus (Table 11): **87.0 % of audited systems sit at one of the two extreme values 0 % or 100 % NaN**, with only 13.0 % distributed across the interior. Any threshold in $(0, 1)$ partitioning the interior in the same way yields essentially the same flagged set; $\tau_{A7} = 0.5$ is reported for parsimony and is robust to sensitivity sweeps in the tested range.

The empirical distribution of ρ_s^{A7} and the per-class incidence on the audited corpus are reported in Section 2.7 (Results, Table 11).

4.7 Multi-source hybridisation

The output of the purging stage enriches each station with a contextual profile obtained by articulating six reference sources (Table 15). Hybridisation is performed either by spatial joining (point-in-polygon, buffer queries) or by administrative aggregation (*i.e.*at the IRIS or municipal level), depending on the granularity of the source. Each row of Table 15 corresponds to one or more typed columns of the released parquet schema (Table 16).

Table 15: Reference sources used for contextual enrichment. Every listed source contributes at least one column to the released parquet schema (Table 16). Out-of-scope sources considered during prototyping but not retained in the v1.0 release (OpenStreetMap point-of-interest density, FUB cycling barometer score) are documented in the audit report.

Source	**Granularity**	**Variables**	**Operation**
INSEE Filosofi	IRIS / municipality	Median income/CU, local Gini, decile, car-less share	Aggregation
INSEE Recensement	municipality	Share of commute by bike (`part_velo_travail`)	Aggregation
IGN BD ALTI	DEM raster	Elevation, topographic roughness index	Spatial samp
IGN BD TOPO	polylines	Cycle-infrastructure linear (300 m buffer)	Spatial join
ONISR BAAC (Cerema)	geolocated accident	Severe-crash count (500 m, 5 yr)	Spatial join
National GTFS aggregator	GTFS stops	Heavy-transit stops within 300 m	Spatial join

4.8 Released schema

The Audit Catalogue Parquet exposes 46 columns documented in Table 16: 14 from the raw GBFS payload (preserved verbatim or normalised), 11 from the audit pipeline (typed `station_type` enum, capacity-raw and capacity-audited pair, seven per-station anomaly flags A1–A7, operator label, and audit-confidence rating), 5 from intra- and inter-system network geometry, 4 from spatial-equity hybridisation, 3 from infrastructure hybridisation, and 2 from multimodal-access hybridisation. Every column carries a stable name, a declared dtype, a source pointer and a completeness rate; the machine-readable equivalent is shipped as a JSON Schema, a Frictionless Data Package descriptor and a Croissant manifest. The eleven audit-pipeline columns are the key novelty of the catalogue: they make the audit's verdict inspectable per row, rather than aggregated in a separate report.

Empirical completeness on the 5,442 dock-based stations is above 93 % for every contextual variable except the optional `address` field of the raw GBFS payload (Figure 8). The minimum-completeness threshold of 90 % documented in the audit report is therefore reached on every certified column of the released schema.

4.9 Reproducibility infrastructure

The Audit Catalogue is released in alignment with the four FAIR principles [39]. *Findable:* primary deposit on Zenodo under concept DOI `10.5281/zenodo.20125460`, with one DOI per SemVer-tagged release (current 1.0.0, dated 2026-05); mirrored on *Recherche Data Gouv*

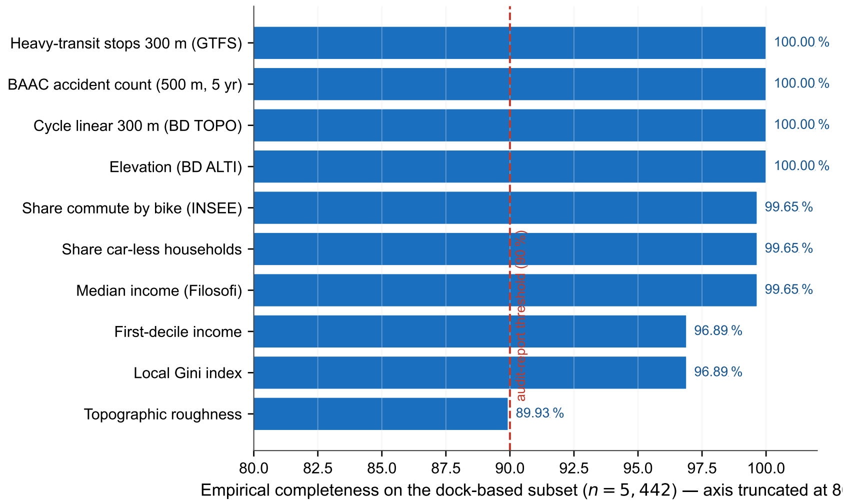

Figure 8: Empirical completeness of contextual enrichment per derived variable on the dock-based subset of the Audit Catalogue ($n = 5{,}442$). All variables exceed the 90 % completeness threshold required by the audit report; missing values remain below 6 % across the corpus.

(HAL/BSO indexing). *Accessible:* Parquet format, direct HTTPS download and Git LFS mirror. *Interoperable:* the schema extends GBFS v3 and is published as a JSON Schema, a Frictionless Data Package descriptor [19], a DCAT-AP record [40] and a Croissant manifest [41]. *Reusable:* released under ODbL v1.0; code under MIT, with audit report and source code. Raw sources are archived with SHA-256 hashes; the Python environment is pinned in `requirements.txt`; a Docker image is published for bit-exact reproduction. The audit pipeline itself is fully deterministic (no random sampling); the only stochastic step is the bootstrap resampling used for the confidence intervals reported in the main text, for which the random seed is pinned at `numpy.random.default_rng(42)`. Idempotence and per-class detector unit tests run in CI on every push; the audit report is regenerated automatically on every release tag.

4.10 Implementation

The released Parquet file is 7.8 MB on disk for the 46,307 stations and loads in under one second into a standard `pandas` session on commodity hardware; typical analytical queries (city-level filtering, bounding-box selection, joins with external panels) complete in tens of milliseconds. The audit pipeline is implemented as a compact Python package (`audit_pipeline`, $\sim$ 560 lines of code across three modules: `core` for the Tier-1 / Tier-2 detectors, `regenerate` for one-command artefact refresh, and the package `__init__` for the public API) with stable function-level entry points documented in the source repository.

Software validation. The audit primitives are accompanied by a unit-test suite (`tests/test_core.py`, 24 tests, 85 % line coverage of the public `audit_pipeline.core` surface) that exercises each of the seven anomaly classes in isolation as well as the combined end-to-end `enrich()` pipeline. Each detector is paired with a deterministic synthetic fixture engineered to trip exactly that class (e.g., a 25-station grid with one transposed-coordinate row for A4, a 32-station fleet spanning more than 300,000 km^2 for A5, a 60 %-NaN-capacity dock-based system for A7) and with specificity controls verifying that healthy fixtures stay unflagged. The suite further checks that the Tier-2 geometry columns (intra- and inter-system k-nearest-neighbour distances, local densities) are consistent with the equirectangular projection used for the meter-scale arithmetic, and that the pipeline is idempotent under repeated application. Tests are executed by `pytest` as a GitHub Actions workflow on every push and pull request, and locally inside the published Docker image.

We stress that this suite validates the correspondence between the implemented logic and the published anomaly taxonomy; it is not a substitute for the hand-curated cross-operator audit, which provides the external ground truth against which the pipeline's verdicts were calibrated.

5 Data and code availability

Lead contact. Requests for further information should be directed to and will be fulfilled by the lead contact, Rohan Fossé (`rfosse@cesi.fr`).

Materials availability. This study did not generate new physical materials. The released artefact is the *GBFS France Audit Catalogue* dataset, a derivative of public open data.

Data and code. The artefacts cover two scopes with explicit licensing boundaries.

- **French Audit Catalogue (Zenodo, ODbL v1.0).** The deposit contains four artefacts under the same concept DOI `10.5281/zenodo.20125460` (current release 1.0.0, 2026-05), mirrored on *Recherche Data Gouv*: (i) the 46,307-station certified Parquet (`stations_gold_standard_final.parquet`, $\sim$ 6.6 MB, 46 typed columns including per-row anomaly flags and audited capacities) ; (ii) the lightweight system-level executive summary (`stations_gold_standard_audit_summary.csv`, $\sim$60 kB, one row per audited system with anomaly counts and operator attribution) for users who want an overview before downloading the full parquet; (iii) the `rejected_stations.parquet` log with exclusion motives; and (iv) the audit report and machine-readable descriptors (JSON Schema, DCAT-AP, Frictionless Data Package, Croissant JSON-LD). The 142 candidate French GBFS feeds inventoried from *transport.data.gouv.fr* and MobilityData are archived with SHA-256 hashes in the same deposit; the certified subset of 123 systems is derived deterministically from them.
- **Source code (MIT licence).** The audit pipeline (`audit_pipeline` Python package, nine-step purging protocol, hybridisation joins, FAIR exporters) is released at `https://github.com/rohanfosse/gbfs-audit-catalogue`, together with the manuscript source, the companion Jupyter notebook (`notebooks/catalogue_recipes.ipynb`) and a focused interactive dashboard with five tabs (Overview, Anomaly browser, Operator audit, Schema, Data explorer) at `app/streamlit_app.py`, deployed publicly at `https://gbfs-audit.streamlit.app`. A `CITATION.cff` file and a Docker image (`ghcr.io/rohanfosse/gbfs-audit:1.0.0`) are provided for bit-exact reproduction. Releases are mirrored to Zenodo via the GitHub–Zenodo integration.
- **Hugging Face mirror.** The released parquet is mirrored as a Hugging Face dataset at `https://huggingface.co/datasets/rohanfosse/gbfs-audit-catalogue` for one-line access via `datasets.load_dataset()`.
- **Global audit reproduction.** The full per-system result of the 1,509-system MobilityData-catalogue audit is versioned as code rather than as data, because the upstream catalogue evolves. A re-run is one command: `python -m audit_pipeline.global_audit` from the source repository. The output (`massive_audit_results.csv` and `massive_audit_summary.json`) is regenerated on every release and archived as a Zenodo asset attached to the corresponding tag.
- **Additional resources.** The 500-station blinded sample used in the internal robustness checks, the orthogonal- and full-task classifier scripts, the $\sigma_{\max}$ sensitivity notebook, and the per-operator characterisation of the French free-floating subset are released as supplementary material alongside the dataset.

A Catalogue use cases

Five concrete reuse patterns are documented below. Each is a self-contained few-line snippet against `stations_gold_standard_final.parquet` ; a complete Jupyter notebook (`catalogue_recipes.ipynb`) is released alongside the dataset.

Use case 1 – Loading the catalogue. A single `read_parquet` call exposes the 46 typed columns; typical project filters use `station_type`, `audit_confidence` and the per-row anomaly flags `flag_A1..flag_A7`. The example below selects only dock-based stations whose audit is unambiguous (zero A1–A7 flags fired); the 40-station gap with the 5,442 dock-based count of Table 2 corresponds to dock-based stations whose single A4 or A6 flag downgraded their `audit_confidence` from *high* to *low*.

```
import pandas as pd
gs = pd.read_parquet("stations_gold_standard_final.parquet")
clean = gs[(gs.station_type == "docked_bike") &
(gs.audit_confidence == "high")]
len(clean)  # returns 5,402
```

Use case 2 – Per-operator anomaly profile. The audit's verdict is exposed at the row level by the seven flag columns, so a single `groupby` returns the per-operator anomaly rates that drive Section 2 :

```
gs.groupby("operator_name").agg(
n=("uid", "size"),
A3_rate=("flag_A3", "mean"),
A7_rate=("flag_A7", "mean"),
).sort_values("n", ascending=False).head(10)
```

The two largest operators on the corpus (*Dott*, $\sim 20\,000$ stations; *Pony*, $\sim 11\,500$) come out at `A7_rate` $= 1.0$ and `A3_rate` $= 1.0$ respectively, which is the row-level expression of the operator-driven hotspots documented in Section 2.

Use case 3 – Raw vs. audited capacity per system. `capacity_raw` preserves the GBFS-declared value (including placeholders and NaN) while `capacity_audited` carries the post-audit count (NaN for non-dock types) ; one can therefore quantify per-system the gap between what GBFS *claims* and what the audit *certifies* :

```
sys = gs.groupby("system_id").agg(
raw_capacity=("capacity_raw", "sum"),
audited_capacity=("capacity_audited", "sum"),
).query("raw_capacity > 1000")
sys["over_count_factor"] = sys.raw_capacity /
sys.audited_capacity.replace(0, 1)
sys.sort_values("over_count_factor", ascending=False).head(5)
```

Use case 4 – Spatial-equity correlation. Per-city correlation between dock-based station count and median income per consumption unit:

```
by_city = (
gs[gs.station_type == "docked_bike"]
.groupby("city").agg(
n=("uid", "size"),
income=("revenu_median_uc", "median"))
)
rho = by_city.n.corr(by_city.income, method="spearman")
# rho ≈ +0.09 on 64 cities (weak)
```

The weak positive coupling at the city level is consistent with the much stronger structural pattern uncovered by use case 5 below: the income-cycling gap is not a city-mean effect but a within-city station-level effect concentrated in the lower income quartile.

Use case 5 – Identifying mobility deserts. Three filters on `station_type`, `revenu_median_uc` and the multimodal `gtfs_heavy_stops_300m` count materialise the desert concept of Section 2 :

```
q1 = gs.revenu_median_uc.quantile(0.25)
deserts = gs[(
gs.station_type == "docked_bike") &
(gs.revenu_median_uc < q1) &
(gs.gtfs_heavy_stops_300m == 0)]
deserts.city.value_counts().head(10)
```

This three-condition filter recovers 1,682 stations (30.9 % of the dock-based corpus) with a Q1 income threshold of 21,040 EUR per consumption unit on the v1.0 release of the catalogue. Figure 9 shows the top fifteen contributing cities; the count is sensitive to the income data vintage and to the GTFS-stop buffer, both of which are documented hyperparameters of the audit.

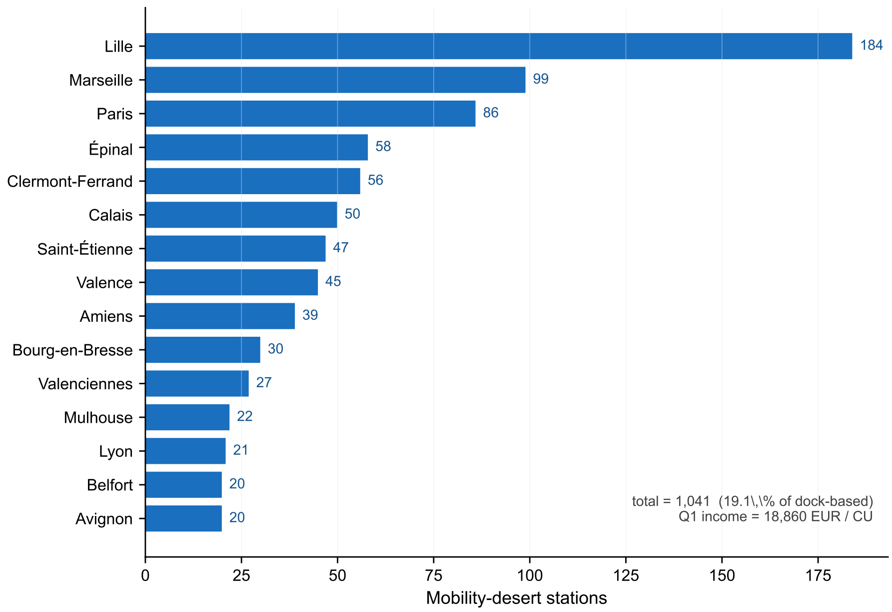

Figure 9: Top fifteen French urban areas by number of dock-based mobility-desert stations, recovered by the three-condition filter of Example 3. A station counts as a mobility desert if its commune sits in the lowest national income quartile and the station has zero heavy-transit stops within a 300 m buffer.

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CRediT authorship contribution statement

Rohan Fossé: Conceptualization, Methodology, Software, Data curation, Formal analysis, Validation, Visualization, Writing – original draft, Writing – review & editing. **Gaël Pallares:** Conceptualization, Methodology, Supervision, Project administration, Writing – review & editing.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used GitHub Copilot and Claude (Anthropic) for code completion and editorial language polishing. After using these tools the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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Table 16: Released schema of `stations_gold_standard_final.parquet`. "Compl." is the empirical completeness rate on the certified subset. Columns marked † are introduced by the audit pipeline; columns marked * are derived from external sources. The `audit_confidence` enum is assigned by `audit_pipeline.core._audit_confidence`: *high* if no A-flag fires, *medium* if a single A3 or A7 flag fires (spec-compliant but consumer-ambiguous), *low* for any other combination of one or more structural flags.

Column	**Type**	**Source / Description**	**Compl.**
Identifiers (5)			
`uid`	string	Audit Catalogue primary key	100 %
`station_id`	string	GBFS native identifier	100 %
`system_id`	string	Operator-system identifier	100 %
`system_name`	string	Operator-system label	100 %
`source`	string	Feed URL / catalogue source	100 %
Spatial and administrative (5)			
`lat, lon`	float64	Geofiltered WGS84 coordinates	100 %
`city`	string	Normalised city name	100 %
`commune_name`	string	INSEE commune label	99.1 %
`code_commune`	string	INSEE commune code	99.1 %
`region_id`	string	Administrative region	100 %
Station description (4)			
`station_name`	string	GBFS station name	99.8 %
`address`	string	GBFS address	41.2 %
`capacity`	int	Declared capacity (may be a placeholder)	96.8 %
`n_stations_system`	int	Total stations in parent system	100 %
Audit pipeline outputs (11)†			
`station_type`	enum	{docked_bike, free_floating, carsharing}	100 %
`capacity_raw`	float64	Raw GBFS capacity (may be NaN or placeholder)	30.1 %
`capacity_audited`	float64	Audited dock count (NaN for non-dock types)	30.1 %
`flag_A1`	bool	Out-of-domain inclusion (carsharing)	100 %
`flag_A2`	bool	Placeholder capacity at system level	100 %
`flag_A3`	bool	Structural over-capacity (free-floating)	100 %
`flag_A4`	bool	Geospatial outlier (3-σ on system centroid)	100 %
`flag_A5`	bool	Out-of-perimeter (system bbox > 50,000 km^2)	100 %
`flag_A6`	bool	Zero-capacity dock	100 %
`flag_A7`	bool	Null-capacity field at system level	100 %
`operator_name`	string	Normalised operator label	100 %
`audit_confidence`	enum	{high, medium, low}	100 %
`fetched_at`	datetime	Timestamp of the audited snapshot	100 %
Network geometry (5)†			
`dist_to_nearest_station_m`	float64	Intra-system KNN distance	99.6 %
`n_stations_within_500m`	int	Intra-system 500 m density	100 %
`n_stations_within_1km`	int	Intra-system 1 km density	100 %
`nearest_system_dist_m`	float64	Distance to any other system	100 %
`catchment_density_per_km2`	float64	Stations per km^2 (1 km buffer)	100 %
*Topography (2)**			
`elevation_m`	float64	BD ALTI elevation	99.4 %
`topography_roughness_index`	float64	Local relief amplitude	98.1 %
*Cycling infrastructure (2)**			
`infra_cyclable_km`	float64	BD TOPO cycle-lane linear (300 m)	97.6 %
`infra_cyclable_pct`	float64	Share of dedicated right-of-way	96.9 %
*Safety (1)**			
`baac_accidents_cyclistes`	int	Severe-crash count (500 m, 5 y)	99.8 %
*Multimodal access (2)**			
`gtfs_heavy_stops_300m`	int	Heavy-transit stops within 300 m	98.7 %
`gtfs_stops_within_300m_pct`	float64	Share of accessible heavy-transit	98.7 %
*Socio-economy (5)**			
`revenu_median_uc`	float64	INSEE Filosofi median income/CU	95.4 %
`gini_revenu`	float64	Local Gini index	93.0 %